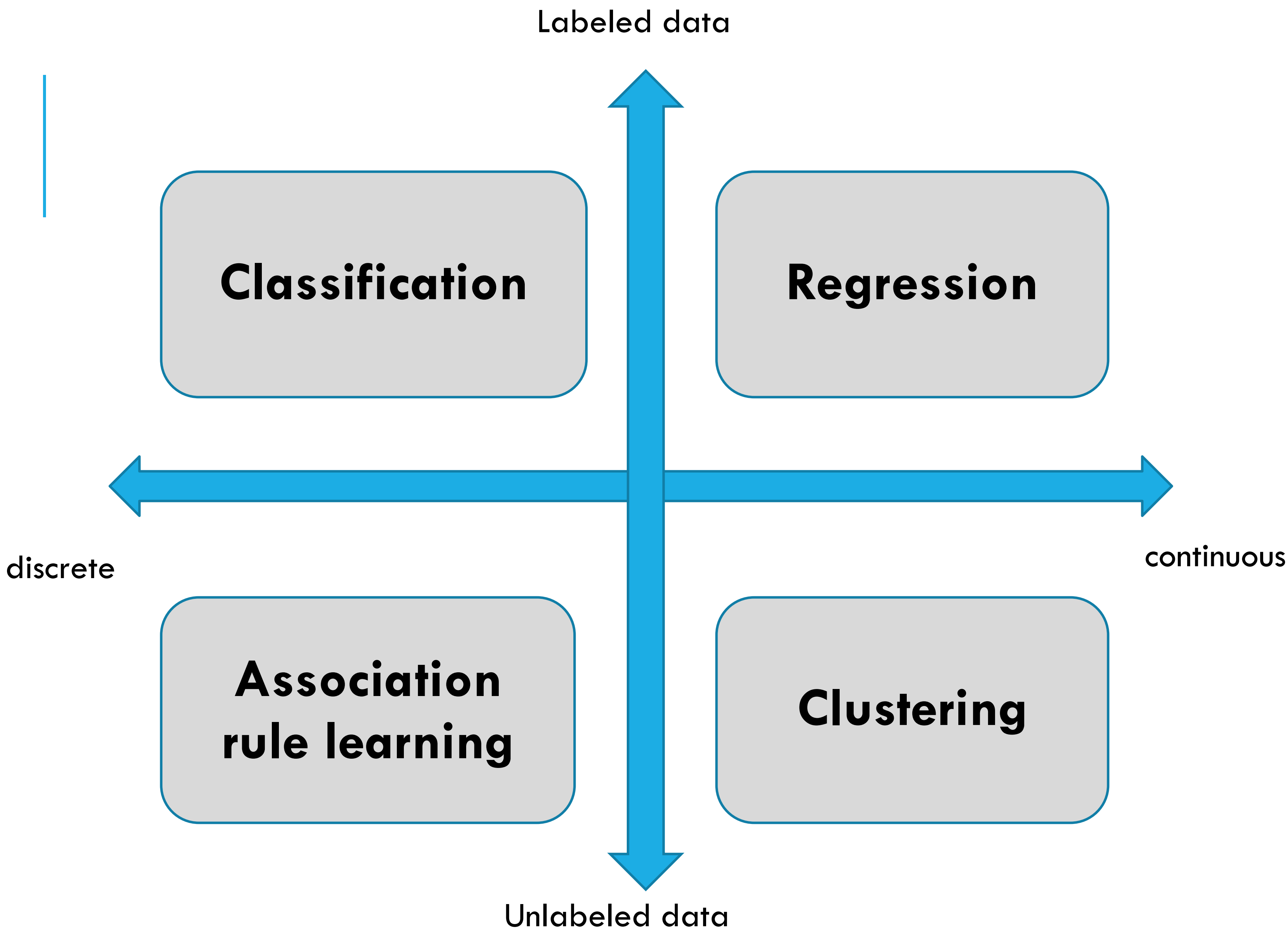
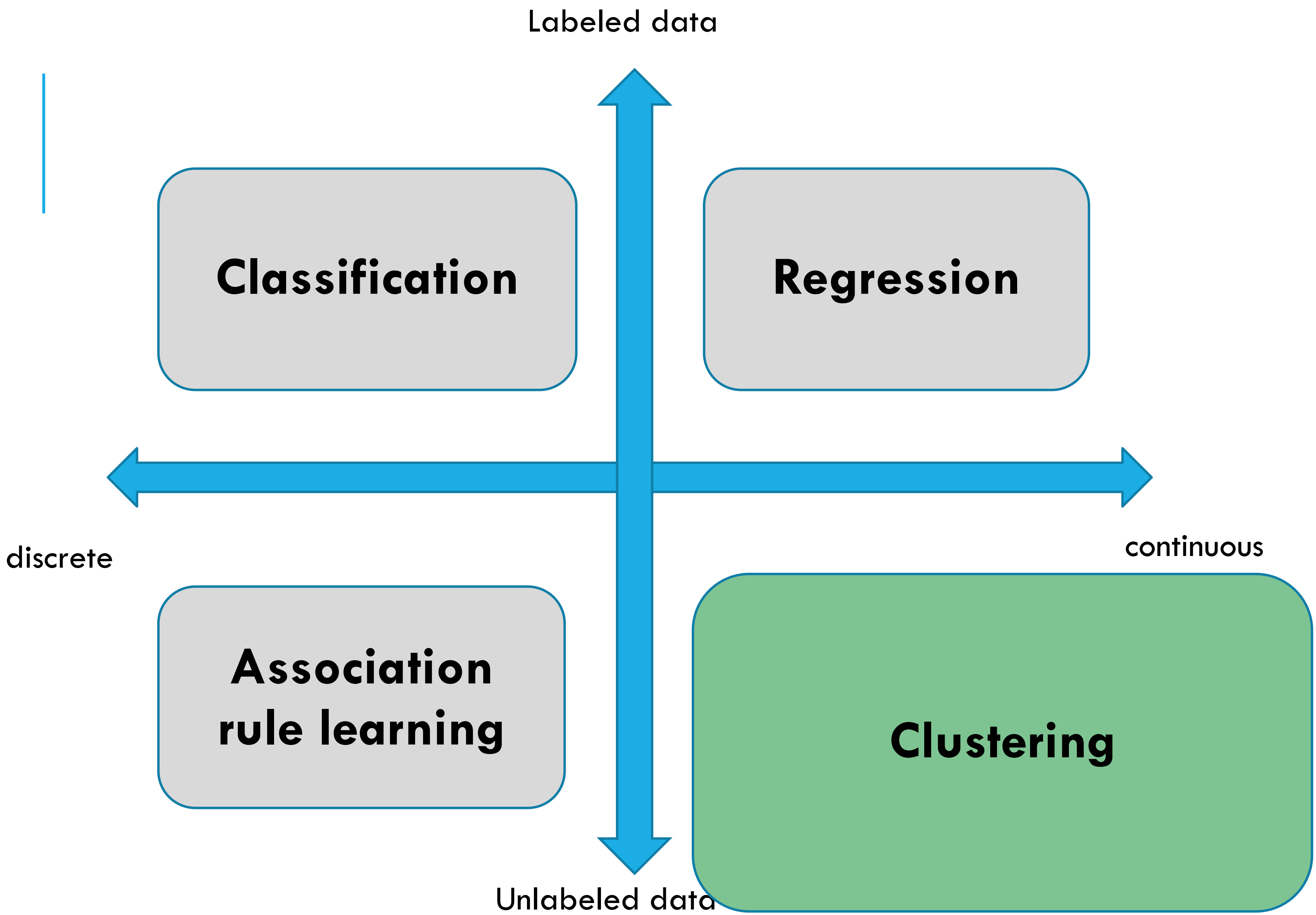
CLUSTERING

Practical aspects in machine learning

Dr Zohar Barnett-Itzhaki


Labeled data
Classification
Regression
discrete
continuous
Association rule learning
Clustering
Unlabeled data


Labeled data
Classification
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Unlabeled data

AGENDA – CLUSTERING PART 1

- Unsupervised learning
- What do we cluster?
- K-means algorithm
- Similarities and distances
- K-means – advanced topics

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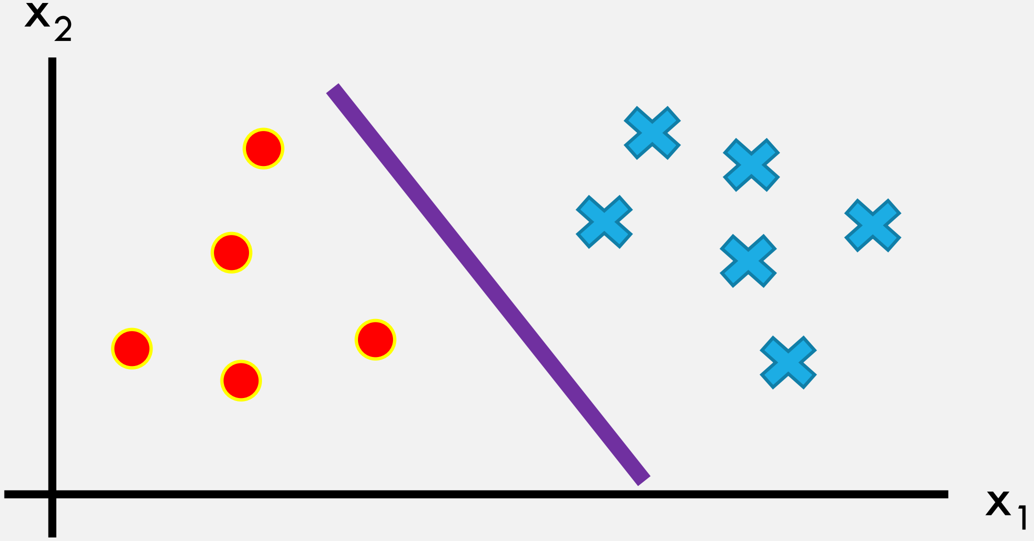
SUPERVISED ALGORITHMS

In Supervised algorithms we have a training set: $\{(x^1,y^1), (x^2,y^2),\ldots,(x^m,y^m)\}$ for which we wanted to find a classifier / separator.

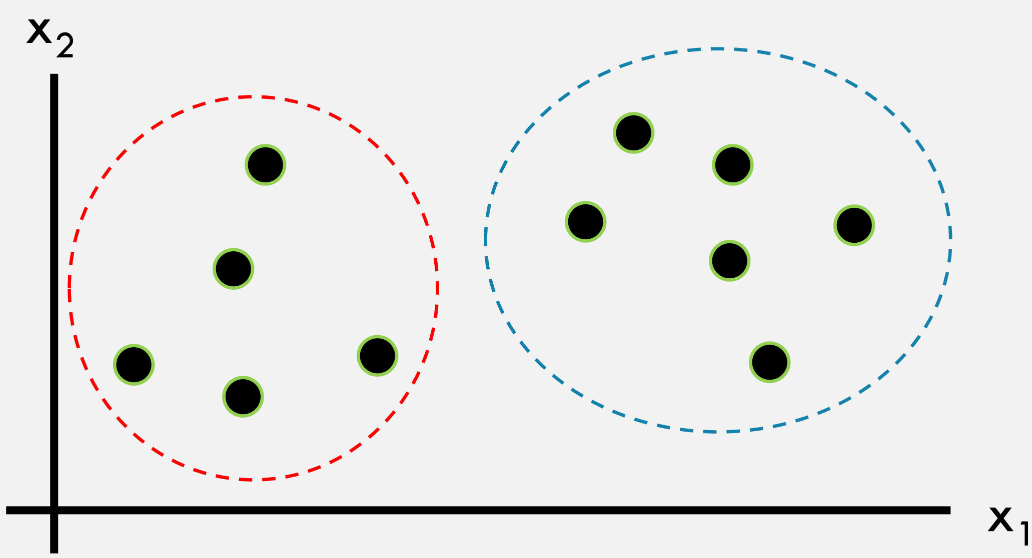
This is a classification problem



UNSUPERVISED ALGORITHMS

In Unsupervised algorithms we have a training set: $\{(x^1), (x^2),\ldots,(x^m)\}$ for which we wanted to find a structure / cluster / rule:

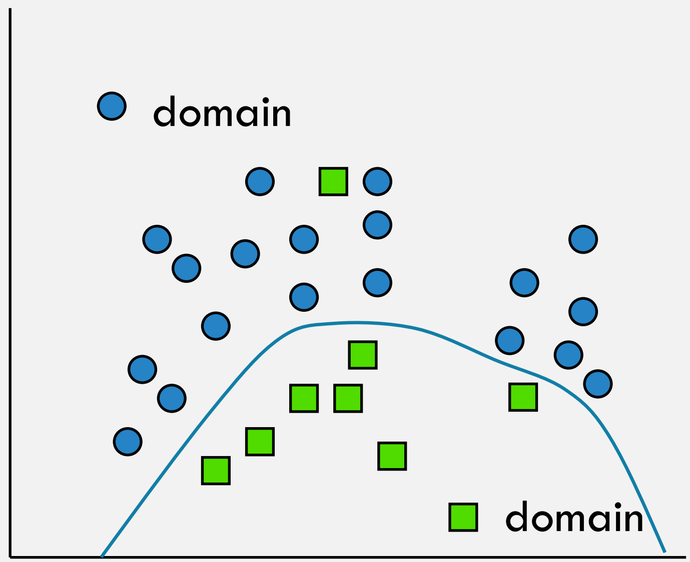
This is a clustering problem



SUPERVISED VS. UNSUPERVISED

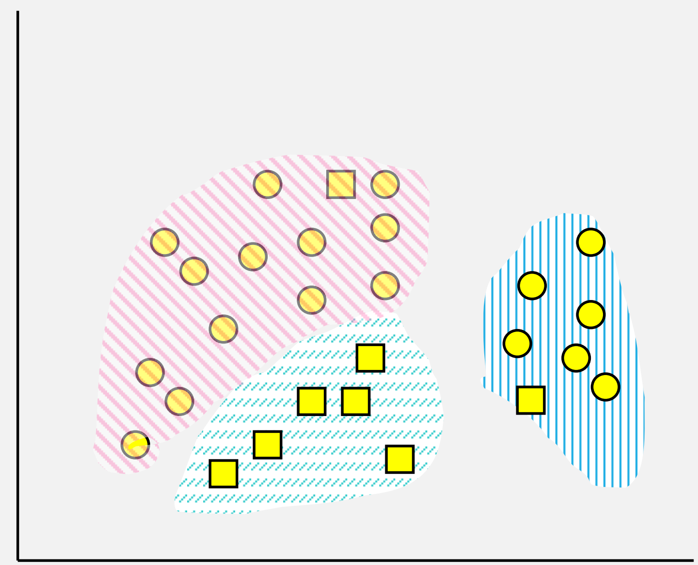
Supervised learning classification

Develop a method to predict the class from pre-labeled instances



Unsupervised learning: clustering

Finds "natural" grouping of instances given un-labeled data



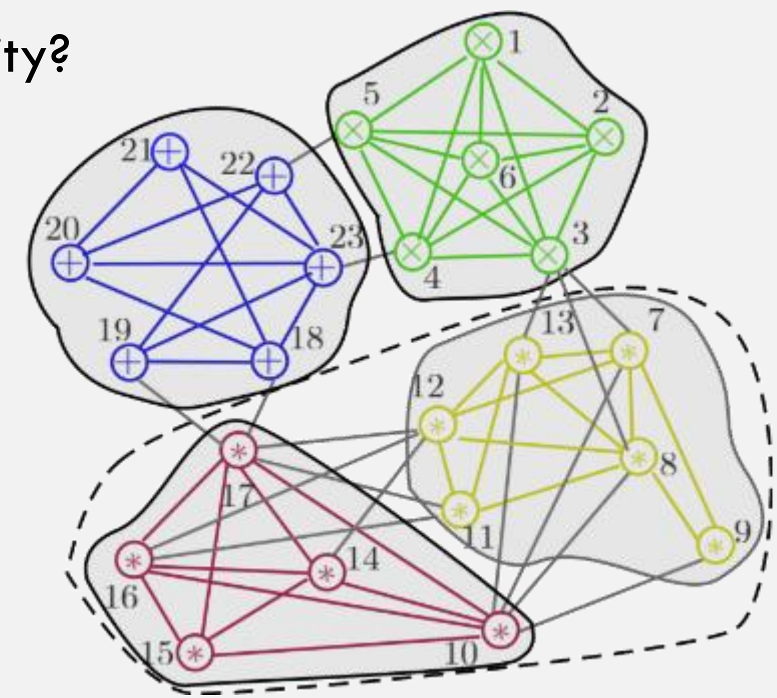
UNSUPERVISED LEARNING – NOTICE!

- There is an inherent issue with unsupervised learning:
- No precise definition of clustering quality
- Clustering is:
  - Application-dependent
  - Ultimately subjective
  - No one solution!

WHAT IS GOOD CLUSTERING?

A good clustering is one when the observations within a group are similar → High intra-class similarity

But what is similarity?



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AGENDA – CLUSTERING PART 1

- Unsupervised learning
- What do we cluster?
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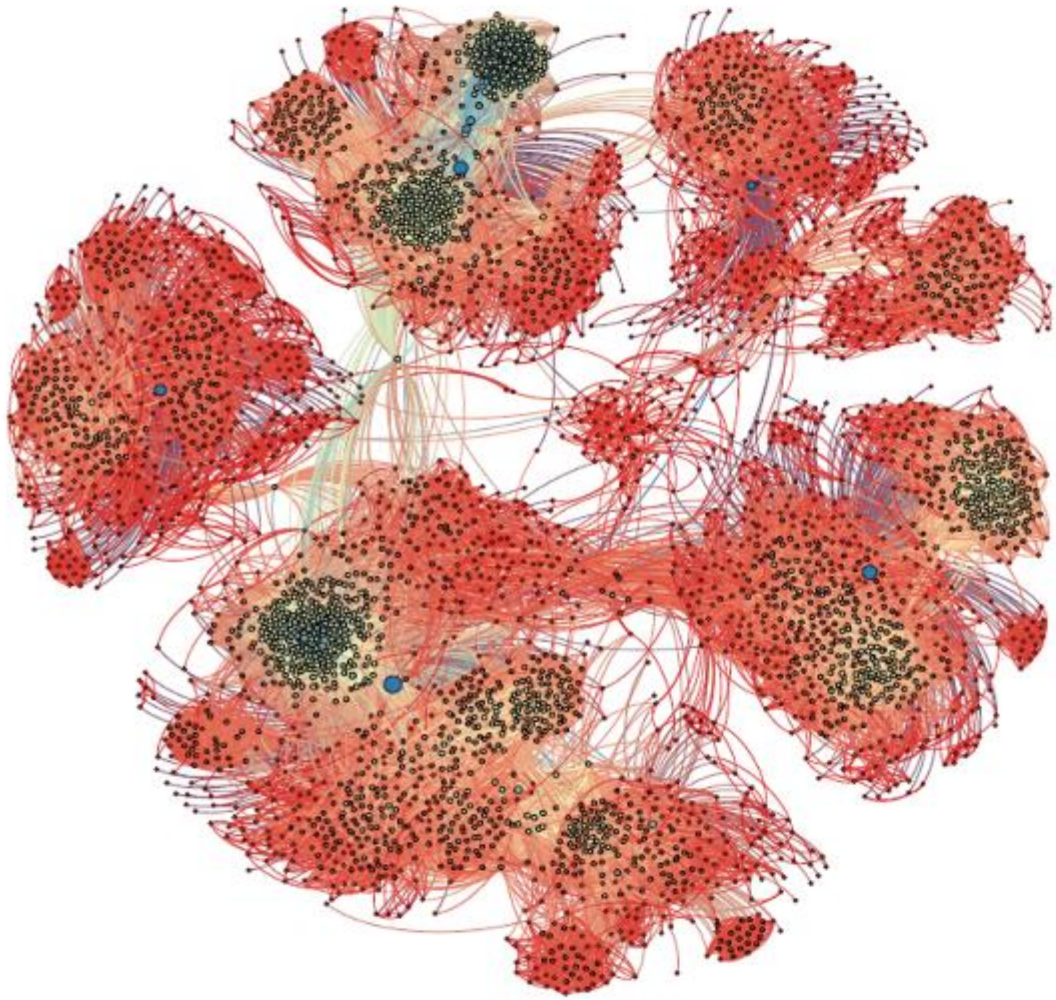
- K-means – advanced topics

WHAT CAN WE CLUSTER?

Social networks

Relationships
Social profile
Professional profiles

SOCIAL NETWORK CLUSTERING



WHAT ELSE CAN WE CLUSTER?

Marketing

Market segmentation
Targeted commercials

WHAT ELSE CAN WE CLUSTER?

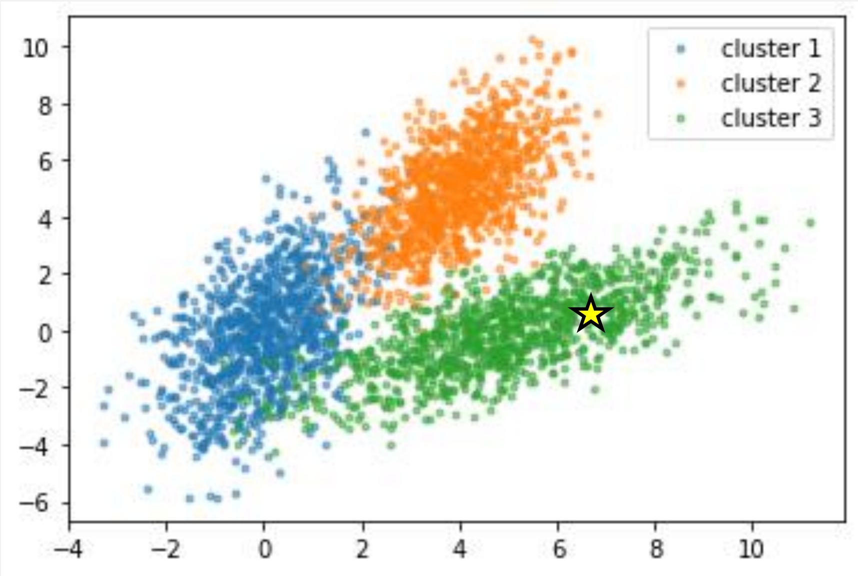
Recommendation systems

Netflix

Amazon

WHAT ELSE CAN WE CLUSTER?

Recommendation systems (Netflix)



Cluster 3 liked:

WHAT ELSE CAN WE CLUSTER?

And much more

Astronomy
Earthquakes

CLUSTERING IMAGES (SEGMENTATION)

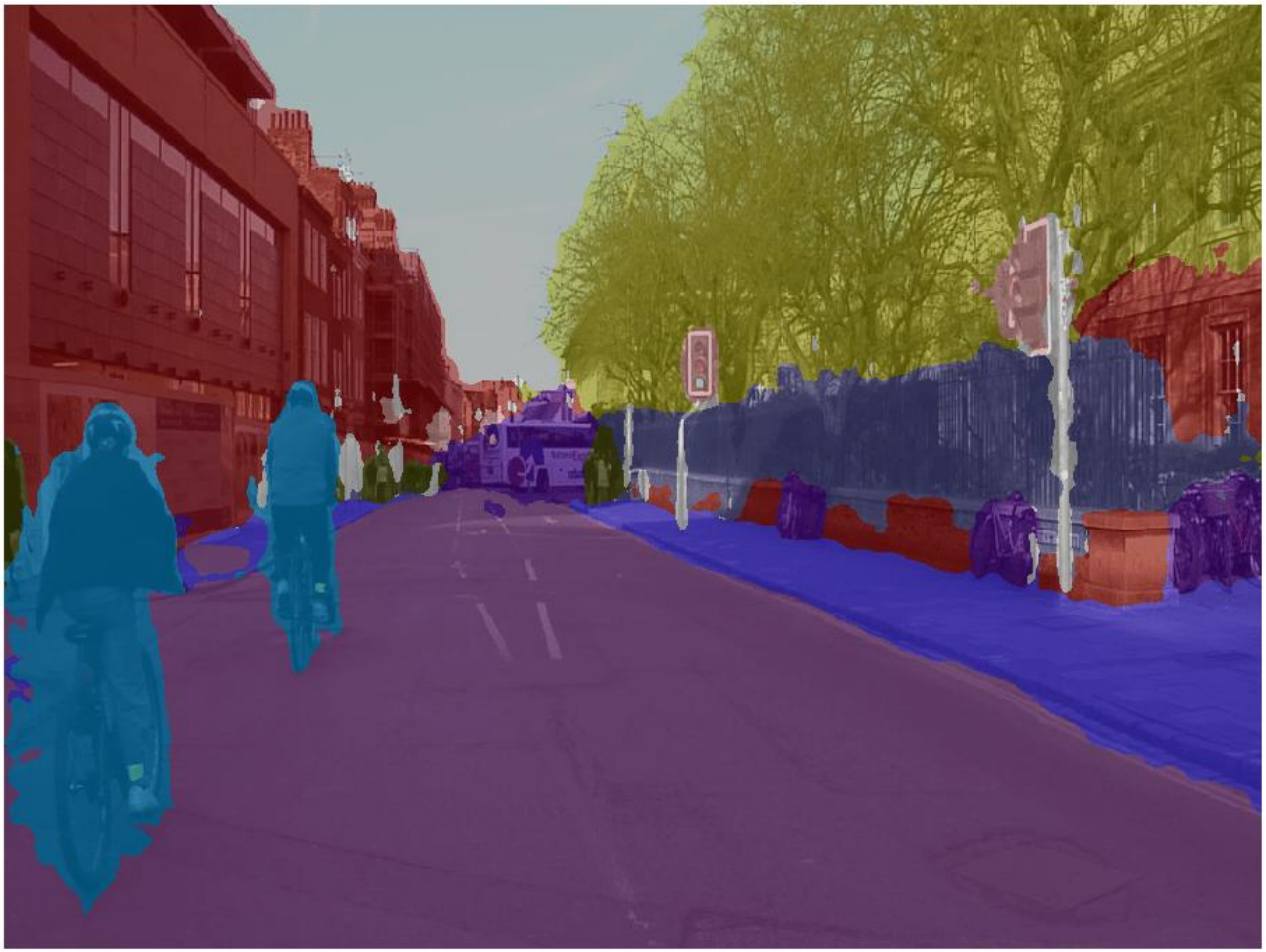


IMAGE CLUSTERING (SEGMENTATION) IS DIFFICULT

Ronen Basri et.al Weizmann Institute

AGENDA – CLUSTERING PART 1

- Unsupervised learning
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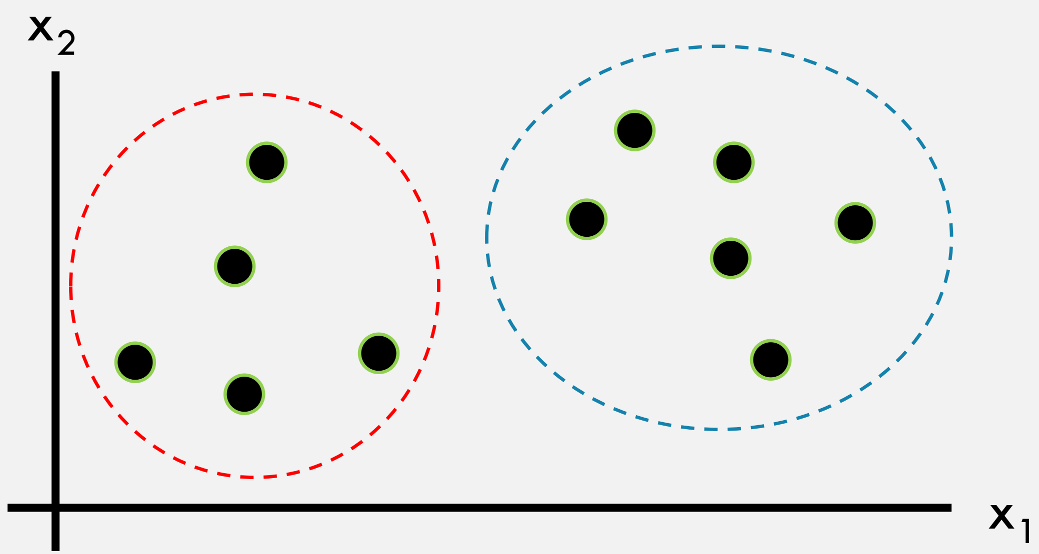
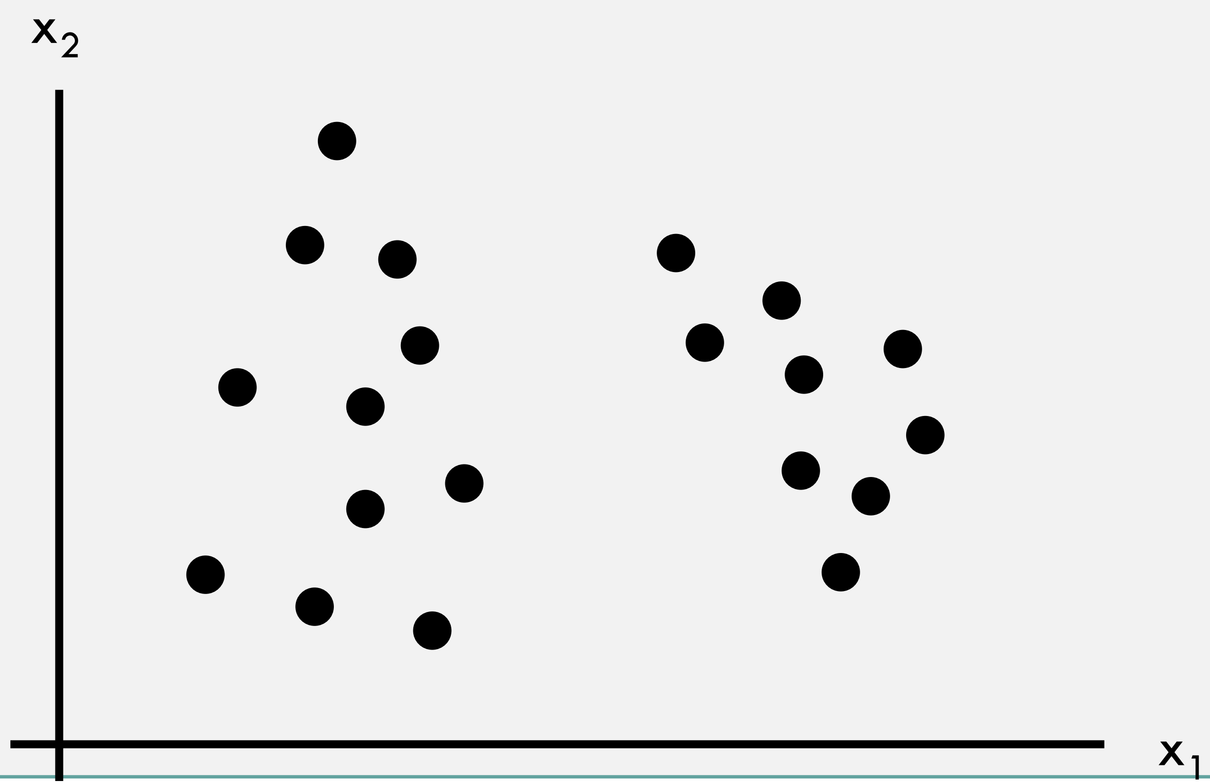
K-MEANS ALGORITHM

- Input: set of samples
- Output: cluster it: split it to meaningful clusters
- The algorithm is iterative (works in iterations)



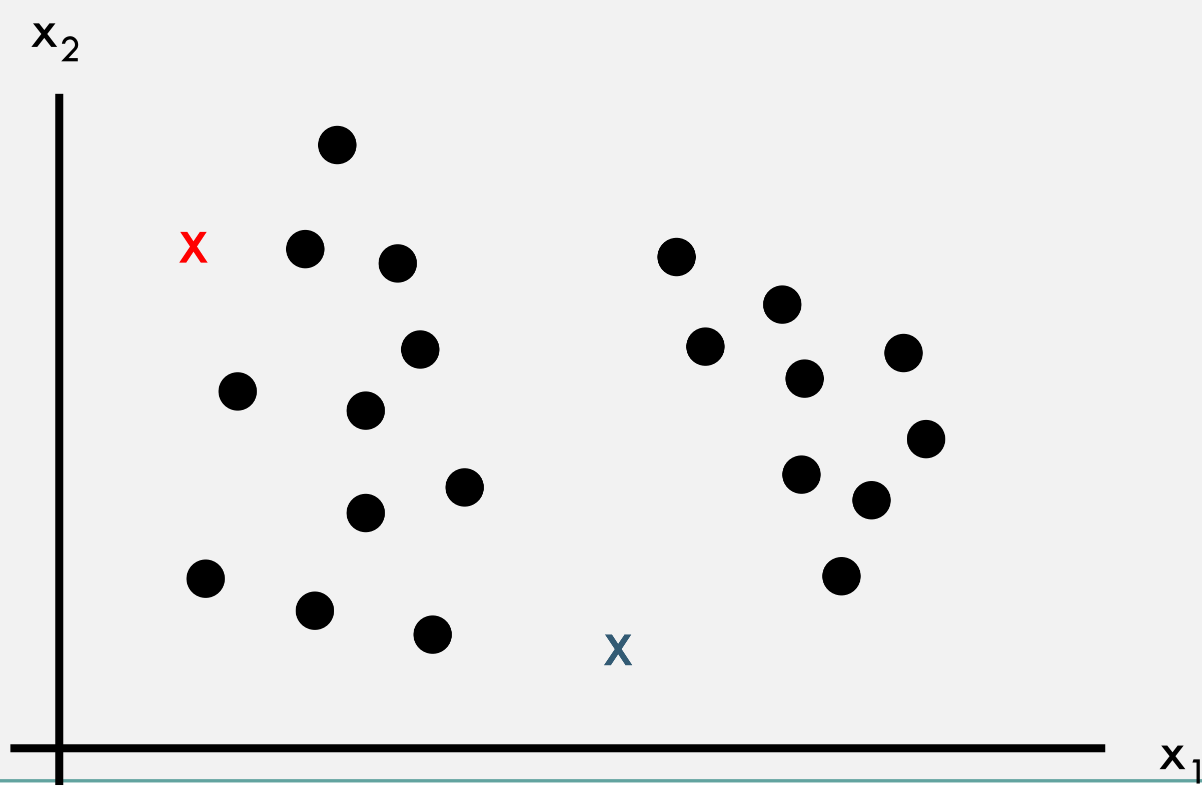
K-MEANS ALGORITHM

- In this example we wish to cluster our samples into two clusters (K=2)



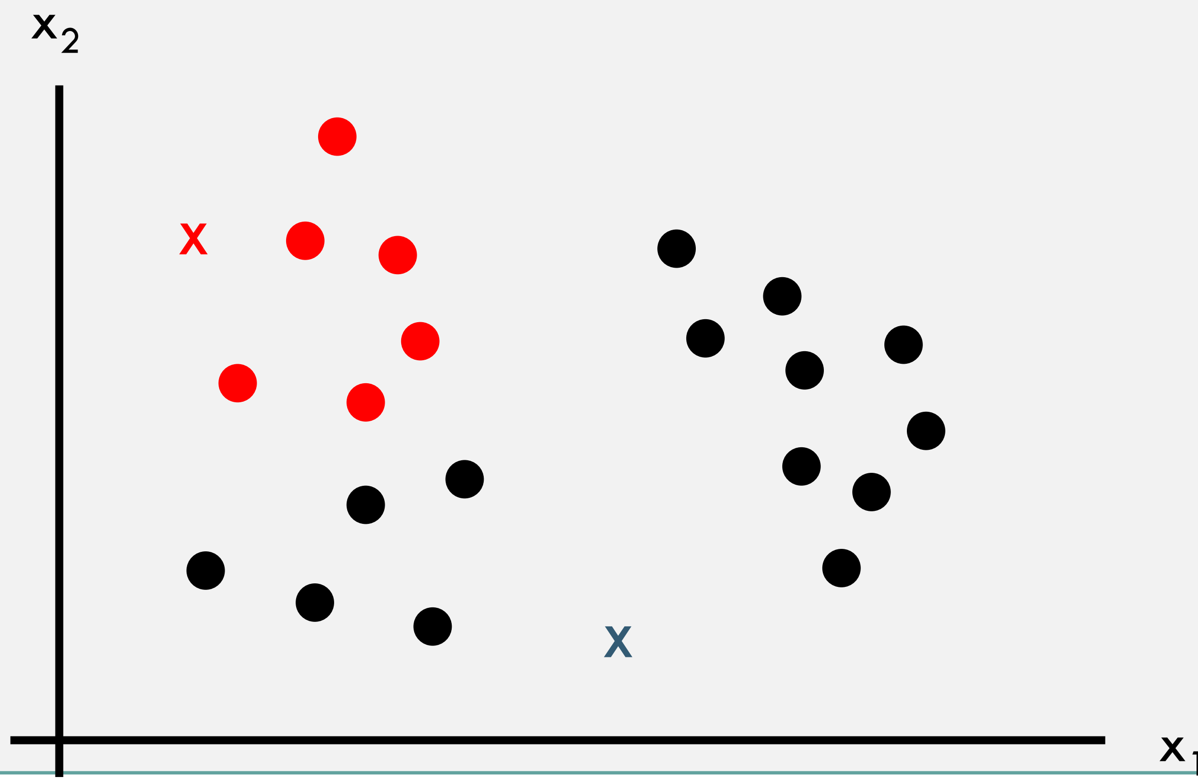
K-MEANS ALGORITHM

- Step 1: choose K random centroids (cores)



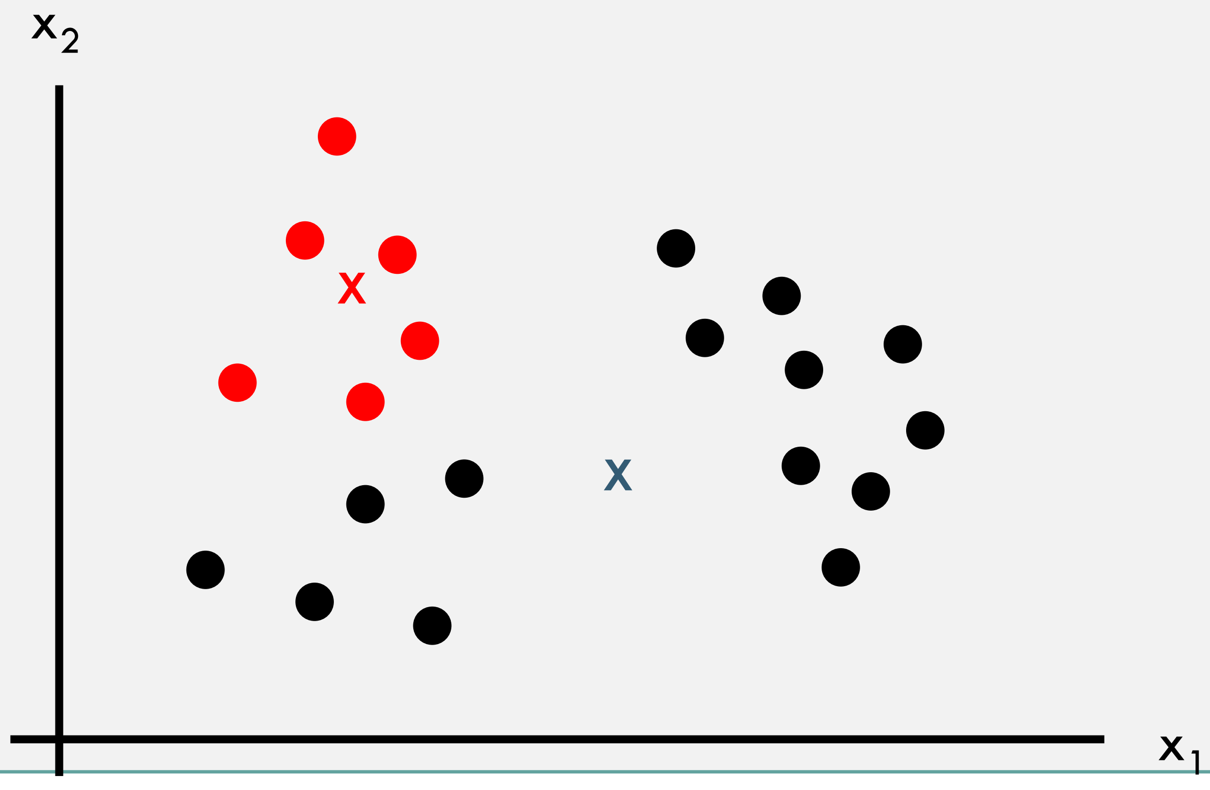
K-MEANS ALGORITHM

- Step 2: assign the centroid color to all samples according to their proximity to the centroid



K-MEANS ALGORITHM

- Step 3: move the centroids to the average locations of the colored samples



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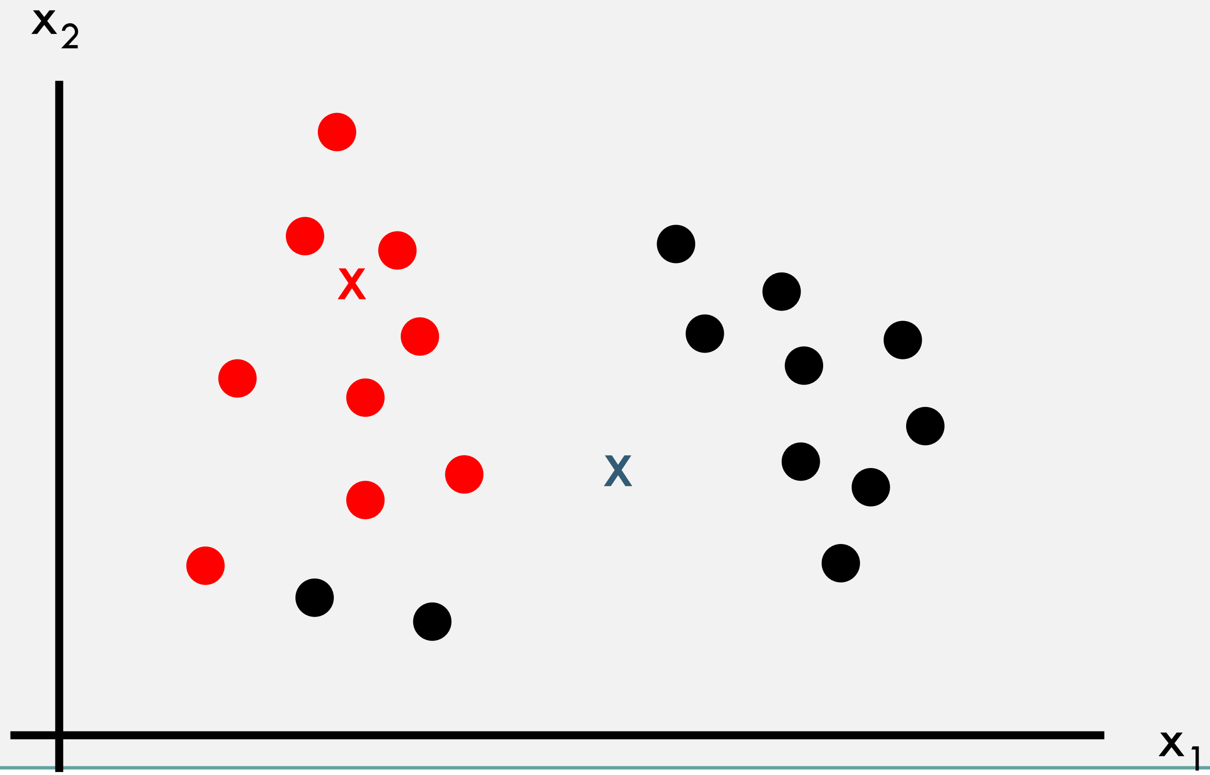
K-MEANS ALGORITHM

- Repeat steps 2 and 3



K-MEANS ALGORITHM

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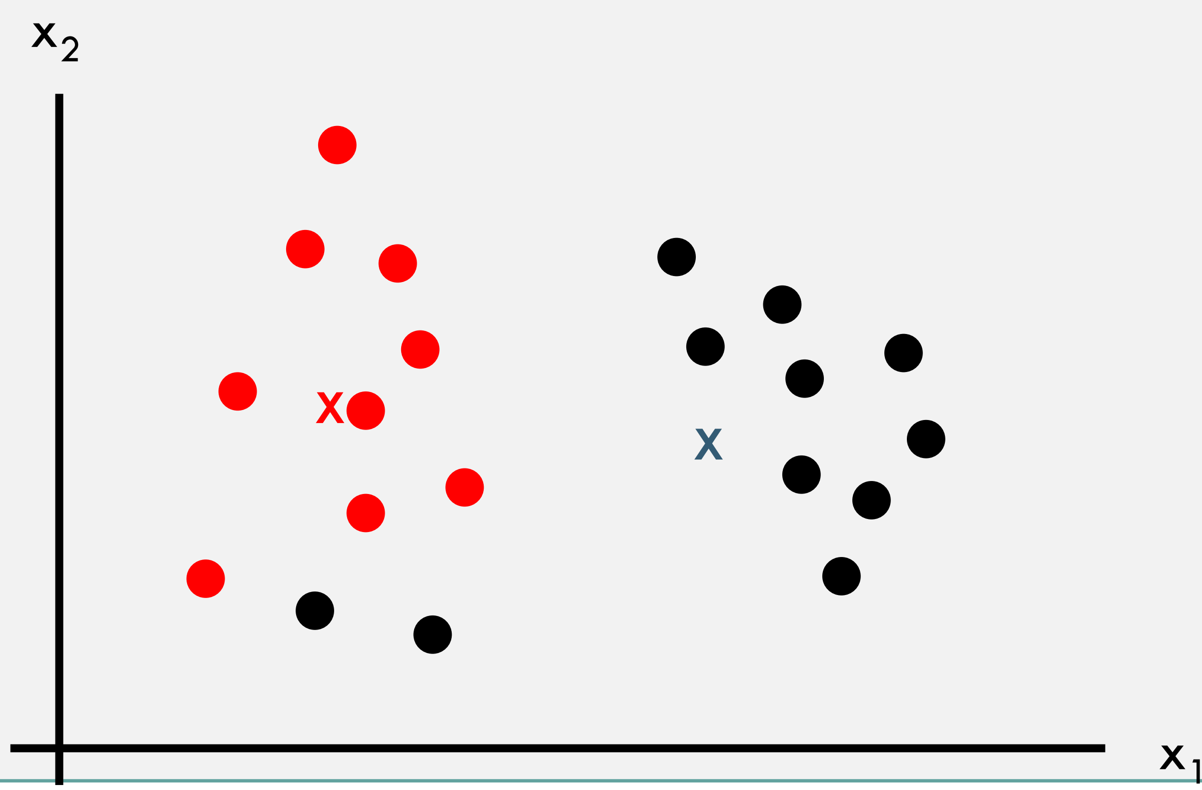

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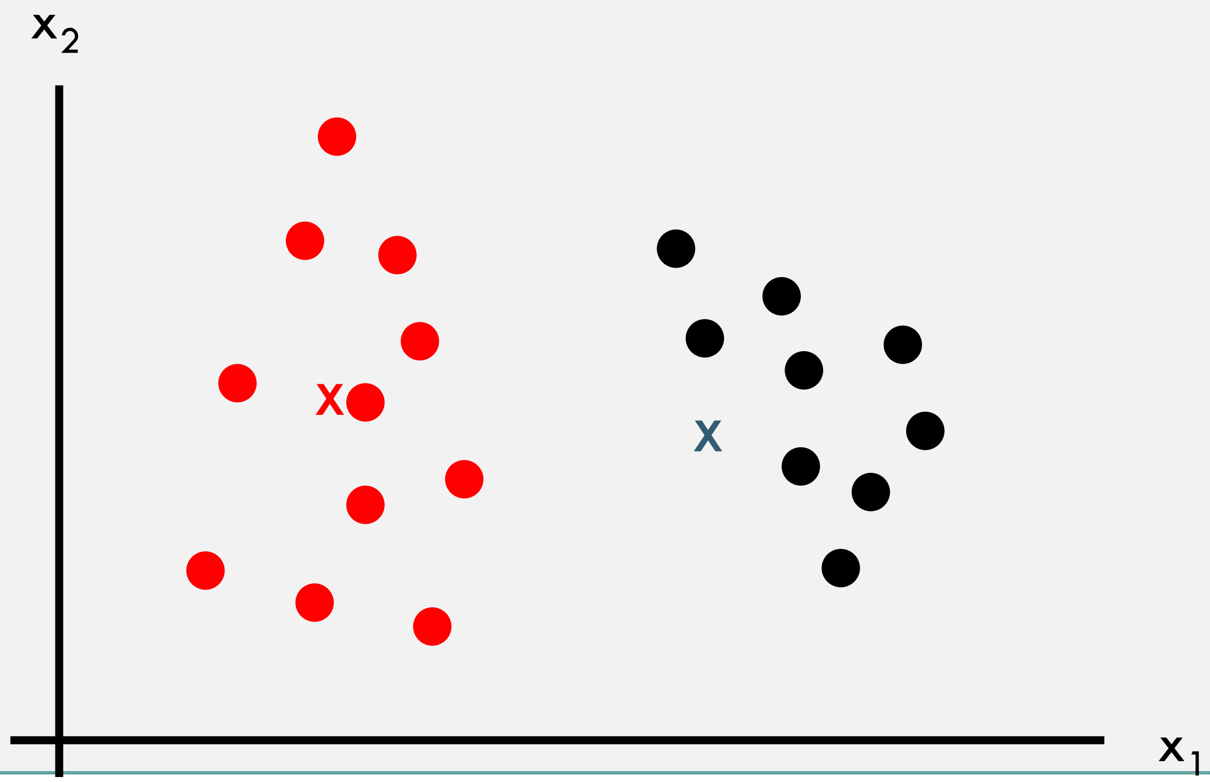
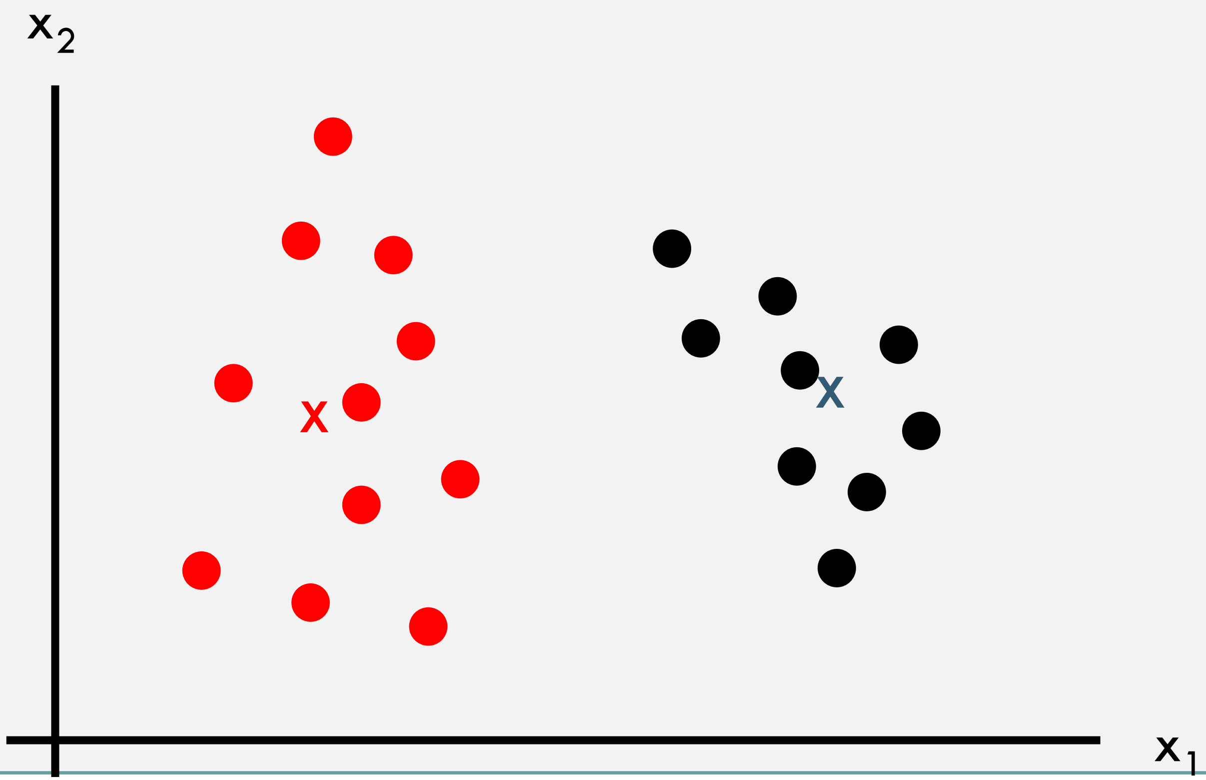


K-MEANS ALGORITHM

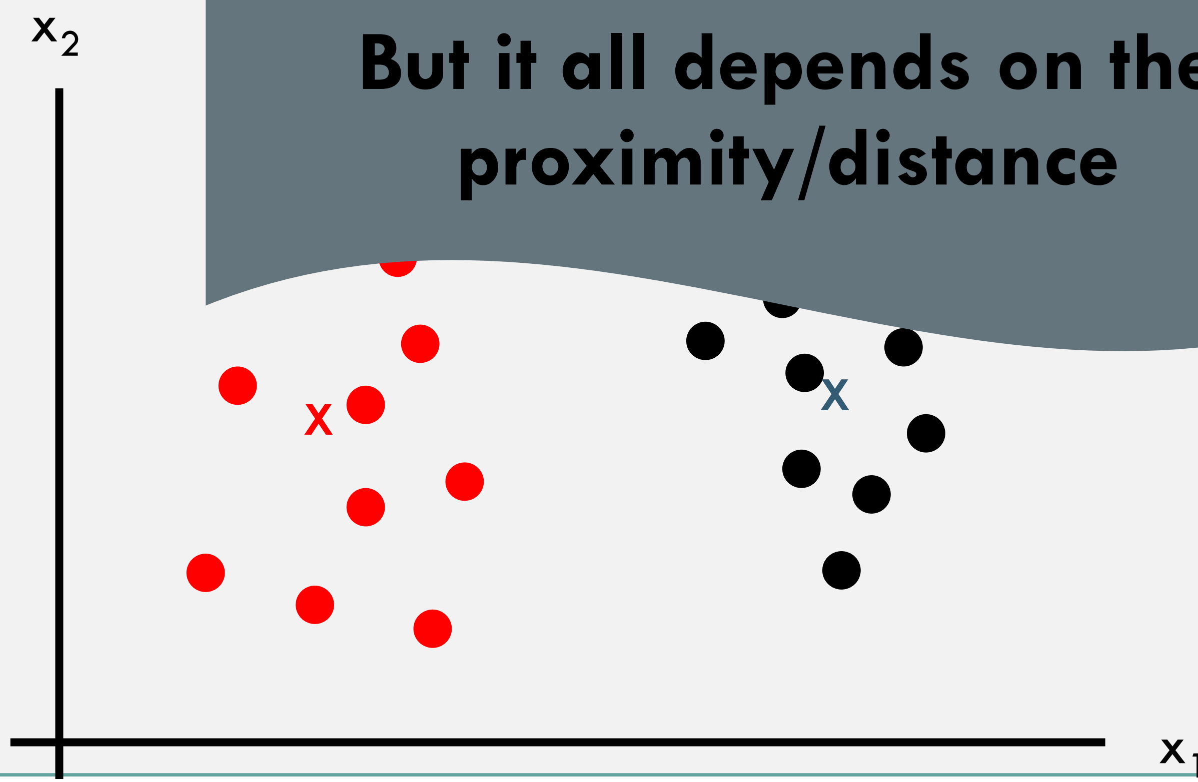
- Repeat steps 2 and 3



K-MEANS ALGORITHM

- Repeat until convergence (the samples don't change their colors)

But it all depends on the proximity/distance



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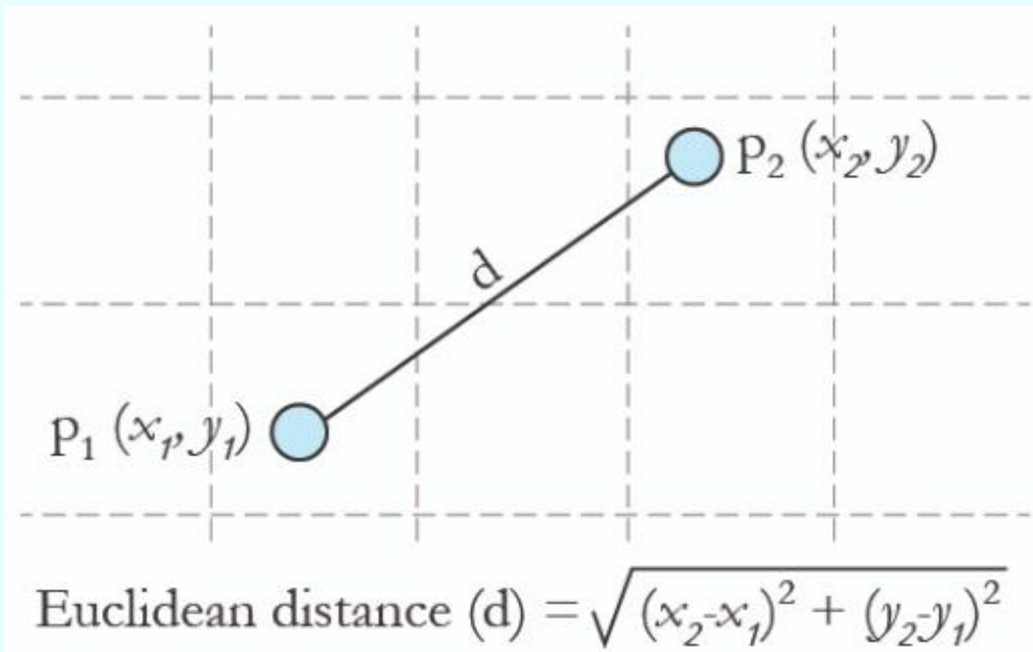
SIMILARITY MATRICES

SIMILARITY AND DISTANCE

- Similarity is the opposite of distance
- If similarity is between 0 to 1
  - → Similarity = 1-distance
- Distance metrics:
  - Euclidean
  - Manhattan
  - 1- correlation
  - Etc..

EUCLIDEAN DISTANCE

- Simplest, fast to compute:



EUCLIDEAN DISTANCE

- Simplest, fast to compute:

$$d(\text{חדרה},\text{תל-אביב})=\sqrt{(|x_{\text{חדרה}}-x_{\text{תל-אביב}}|^2+|y_{\text{חדרה}}-y_{\text{תל-אביב}}|^2)}$$

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EUCLIDEAN DISTANCE

- Simplest, fast to compute:

$$d(a,b) = \|a-b\| = \sqrt{\sum_{i=1}^{N}(a_i - b_i)^2}$$

$$d(a,b) = \sqrt{(a_x - b_x)^2 + (a_y - b_y)^2 + \cdots + (a_n - b_n)^2}$$

$$d(a,b) = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2 + \cdots + (n_a - n_b)^2}$$

$$d(\text{חדרה},\text{תל-אביב}) = \sqrt{(|x_{\text{חדרה}} - x_{\text{תל-אביב}}|^2 + |y_{\text{חדרה}} - y_{\text{תל-אביב}}|^2)}$$

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MANHATTAN DISTANCE

- Inspired by the straight streets of Manhattan
- We do not calculate the aerial distance (Euclidian)
- We look at the full path composed of 90^0 turns

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K-MEANS – MAJOR CHALLENGES

The unassigned centroids and initiation

Local optima

How to choose K?

K-MEANS – MAJOR CHALLENGES

The unassigned centroids and initiation

Local optima

How to choose K?

K-MEANS – UNASSIGNED CENTROIDS

- What can we do in case we have a centroid with no assigned samples?

1) Restart the algorithm
2) Delete the unassigned and use K-1 clusters
3) Move the unassigned to a new random location

- But we can prevent this scenario

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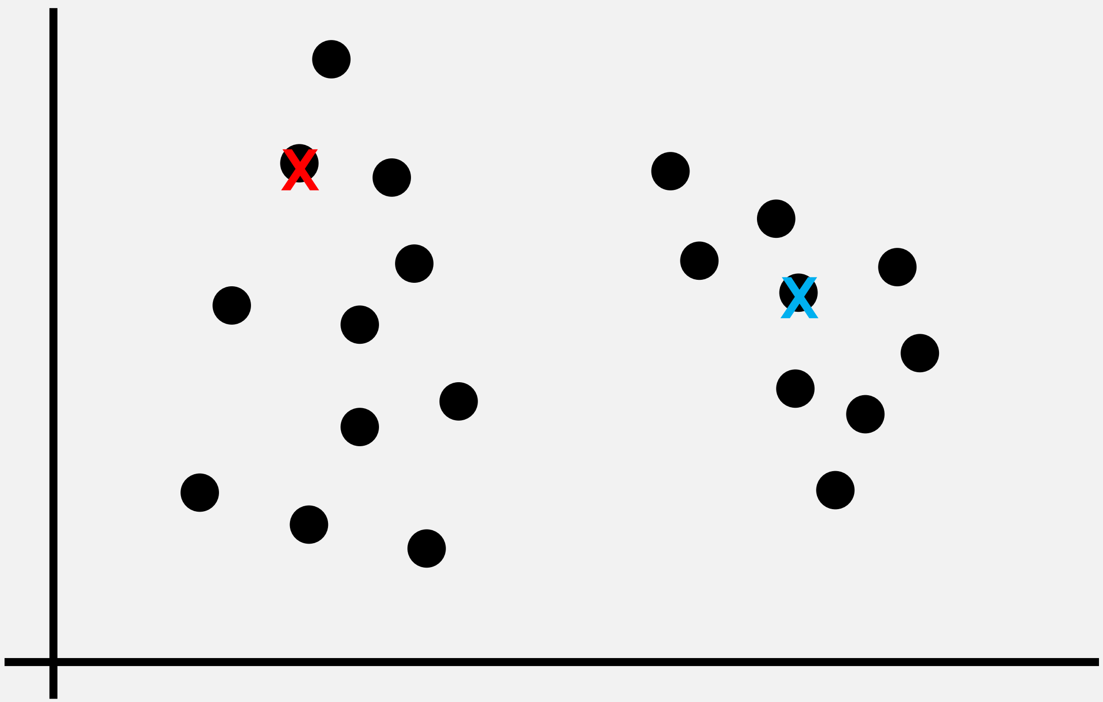
CENTROID INITIALIZATION

Two options:

- Random
- Based on the training set

CENTROID INITIALIZATION

- Randomly choose K samples (K<m) and use them as the centroids



K-MEANS – MAJOR CHALLENGES

The unassigned centroids and initiation

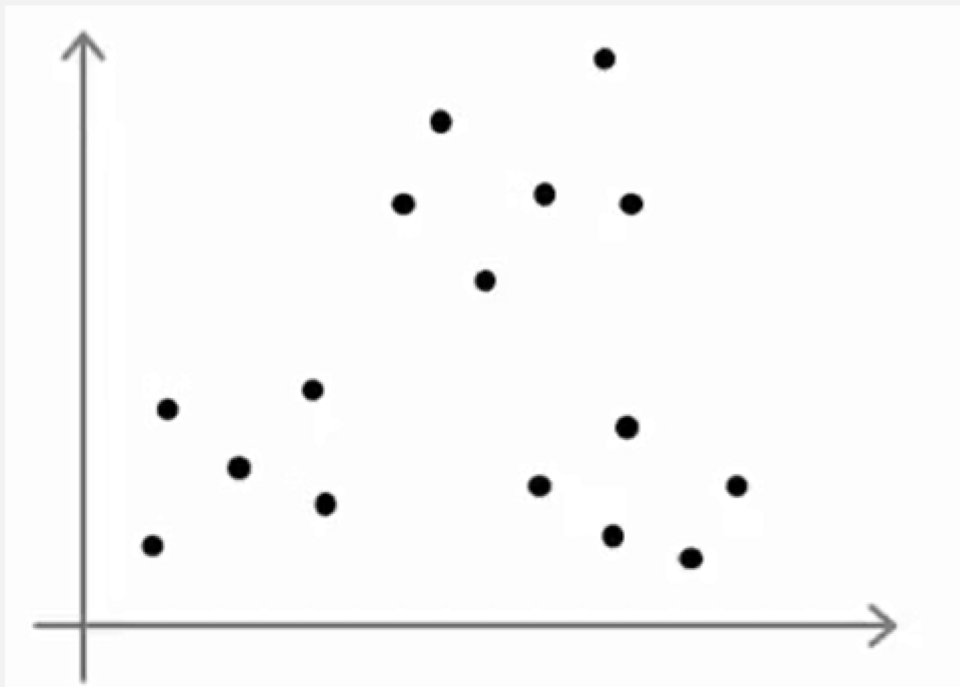
Local optima

How to choose K?

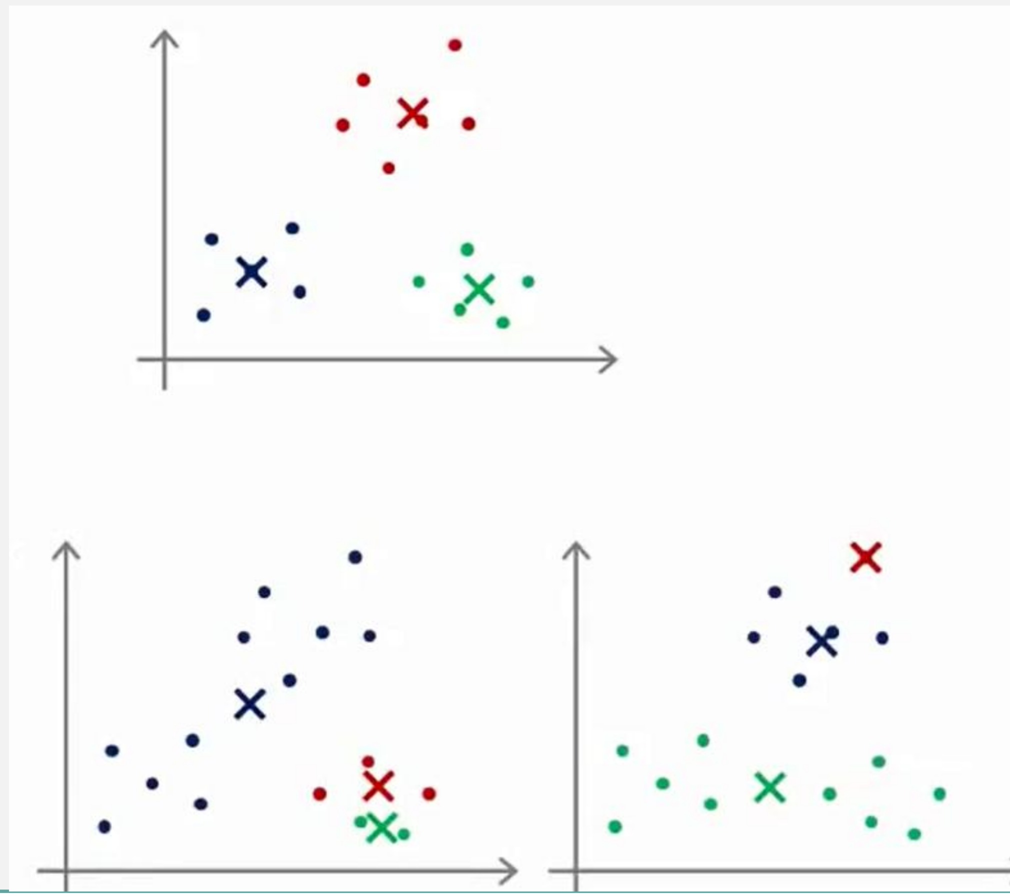
LOCAL OPTIMA

- A major problem in K-means is that the algorithm is not deterministic: specific centroid initializations may result in different clusters
- This may lead to local optima (plural of optimum)

LOCAL OPTIMA

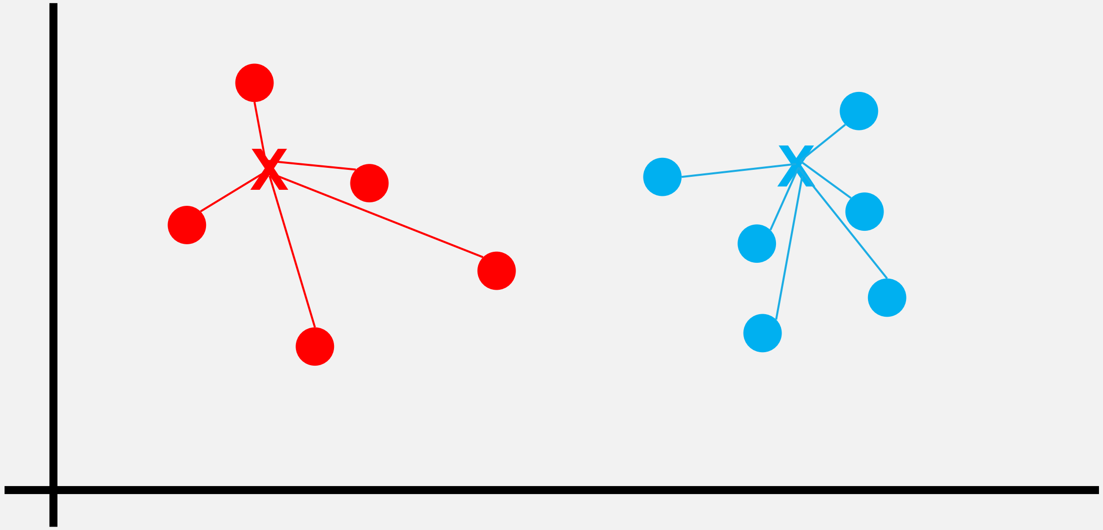


LOCAL OPTIMA



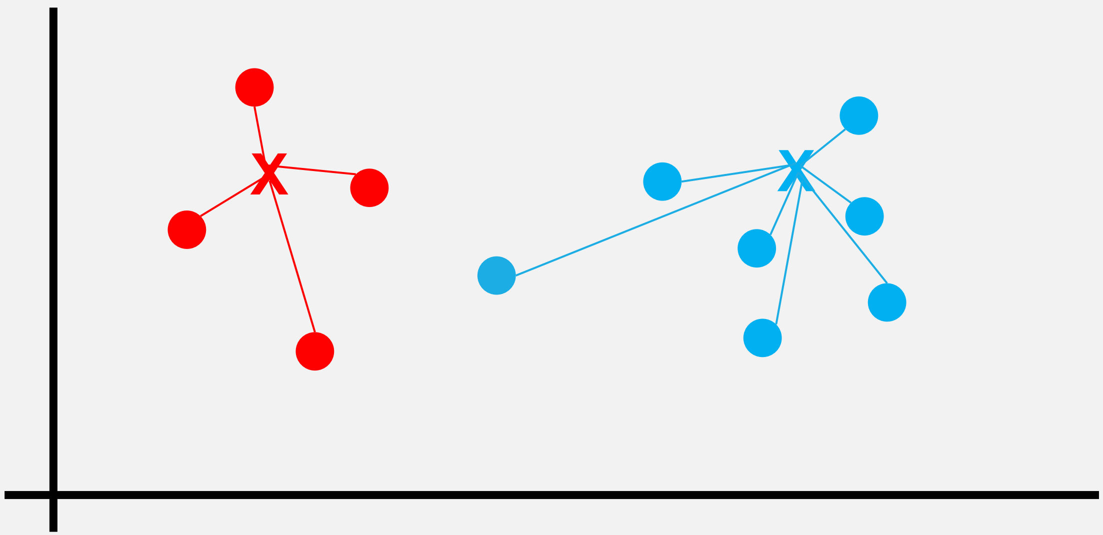
EVALUATION OF CLUSTERING

- In order to evaluate our clustering process we need to calculate the distances of each point from “her” centroid
- This is called “cost function” = average of distances $\Sigma(xi-c)^2/m$
- Of all points from their centroids

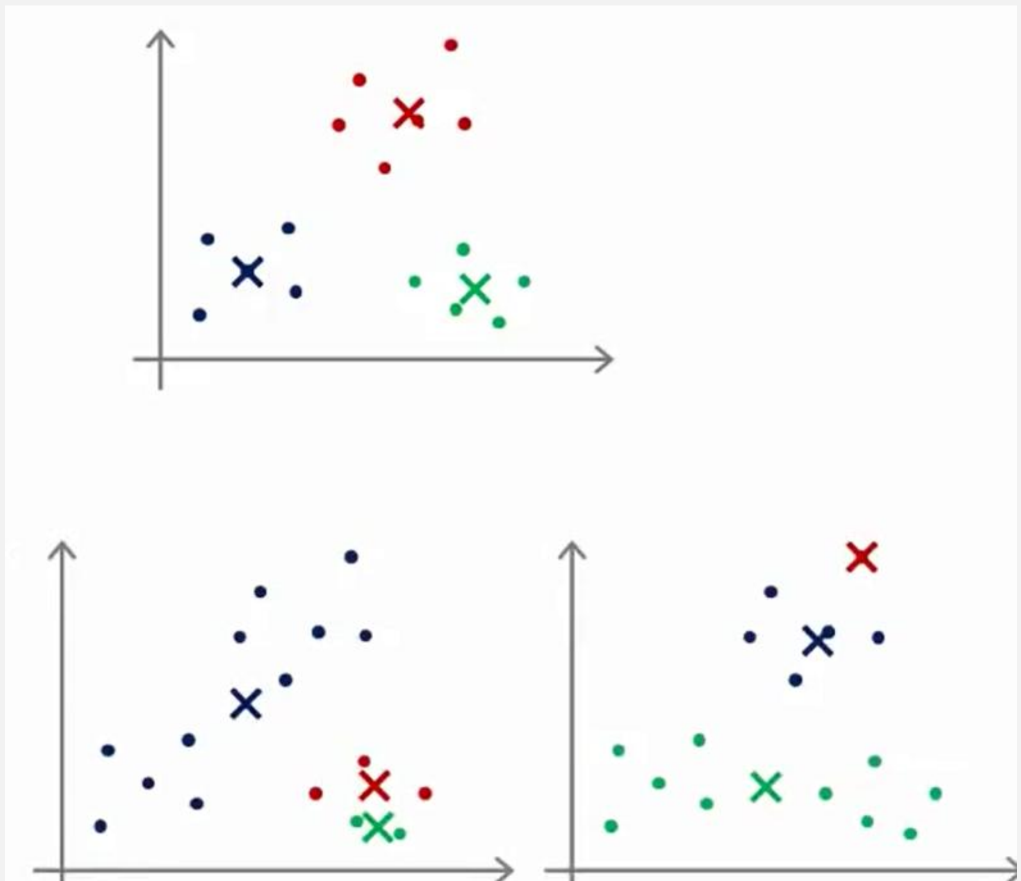


EVALUATION OF CLUSTERING

- An example for not the best clustering

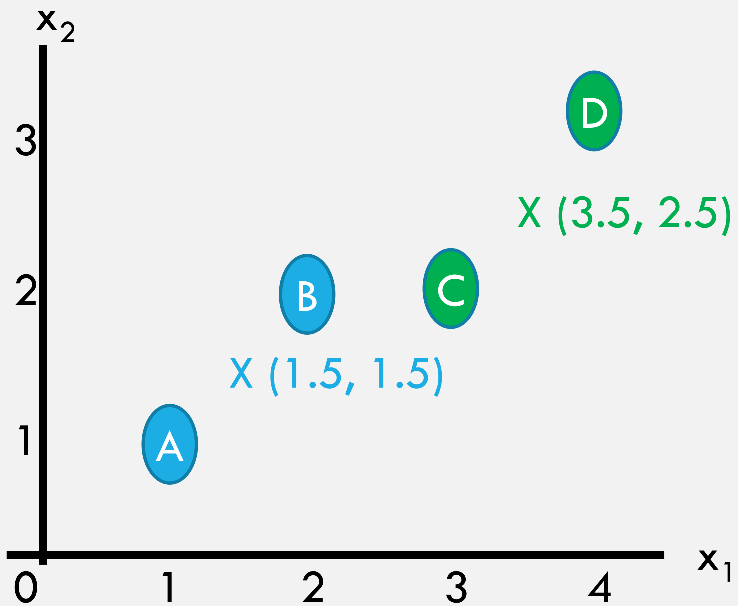


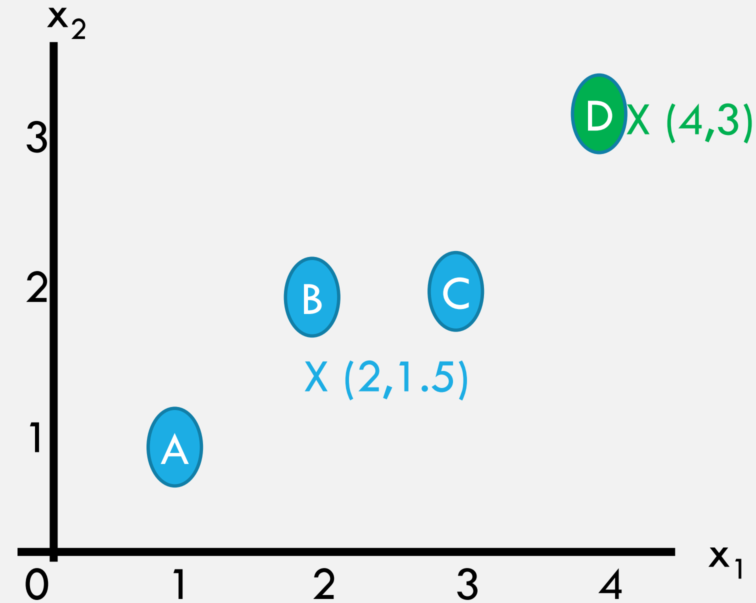
LOCAL OPTIMA



LOCAL OPTIMA - EXAMPLE

- Given the two K-means solutions, what's the best? Why?
- A(1,1), B(2,2), C(3,2), D(4,3)





K-MEANS – MAJOR CHALLENGES

The unassigned centroids and initiation

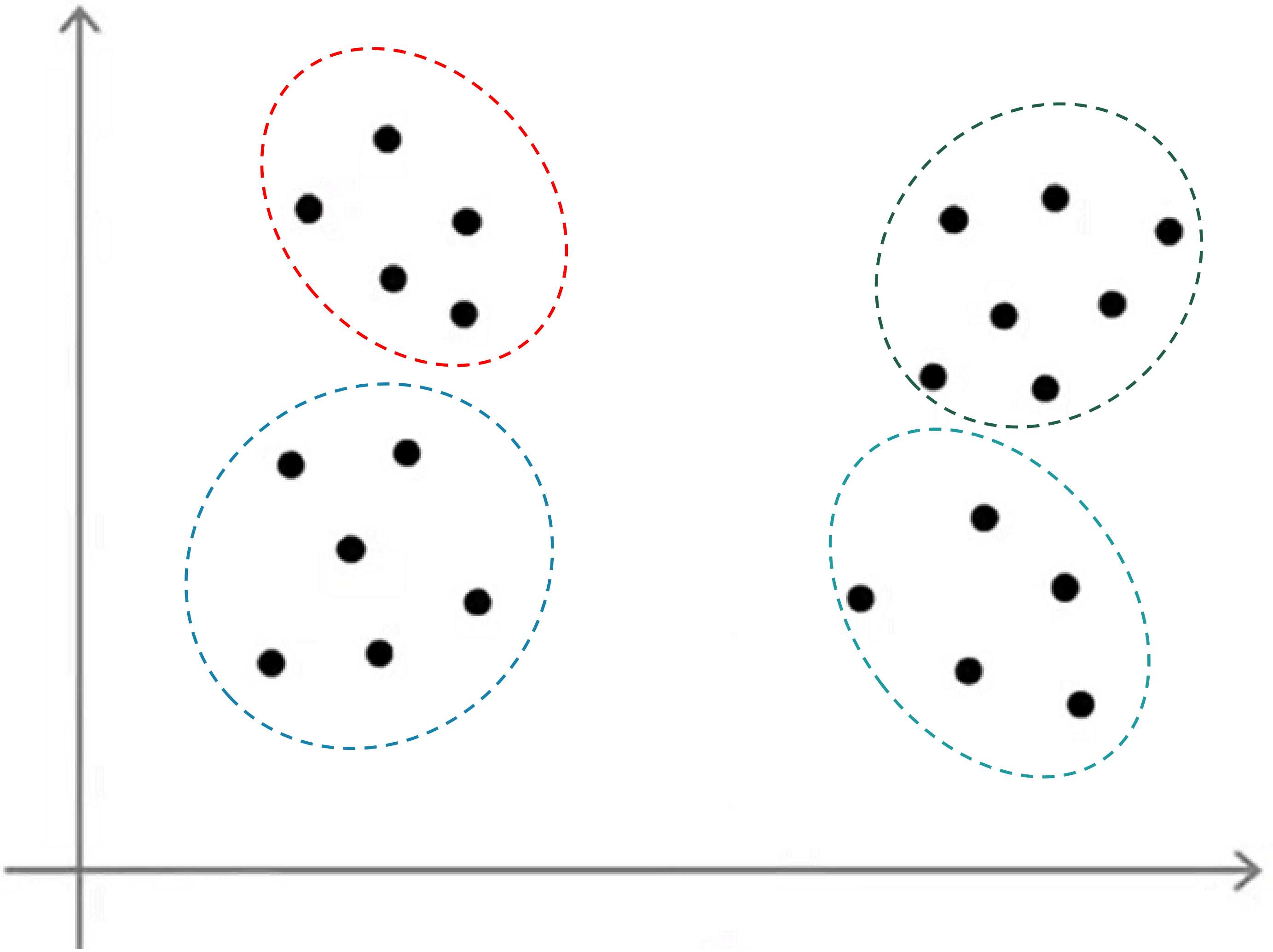
Local optima

How to choose K?

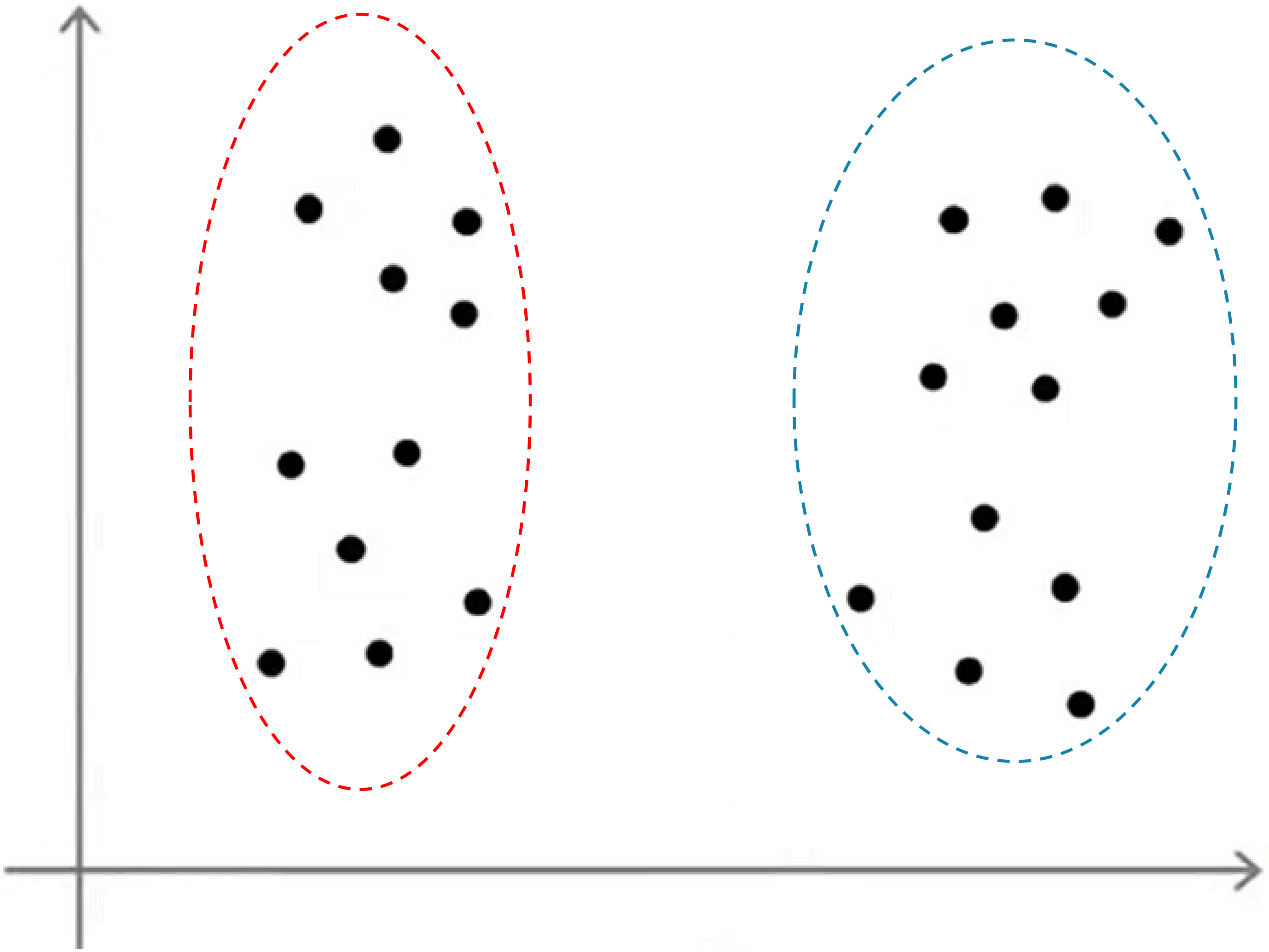
HOW TO CHOOSE THE K?

- This is a nontrivial question
- There are computerized automated approaches to choose K, BUT they don't always yield the best results

WHAT IS THE BEST K IN THIS EXAMPLE?

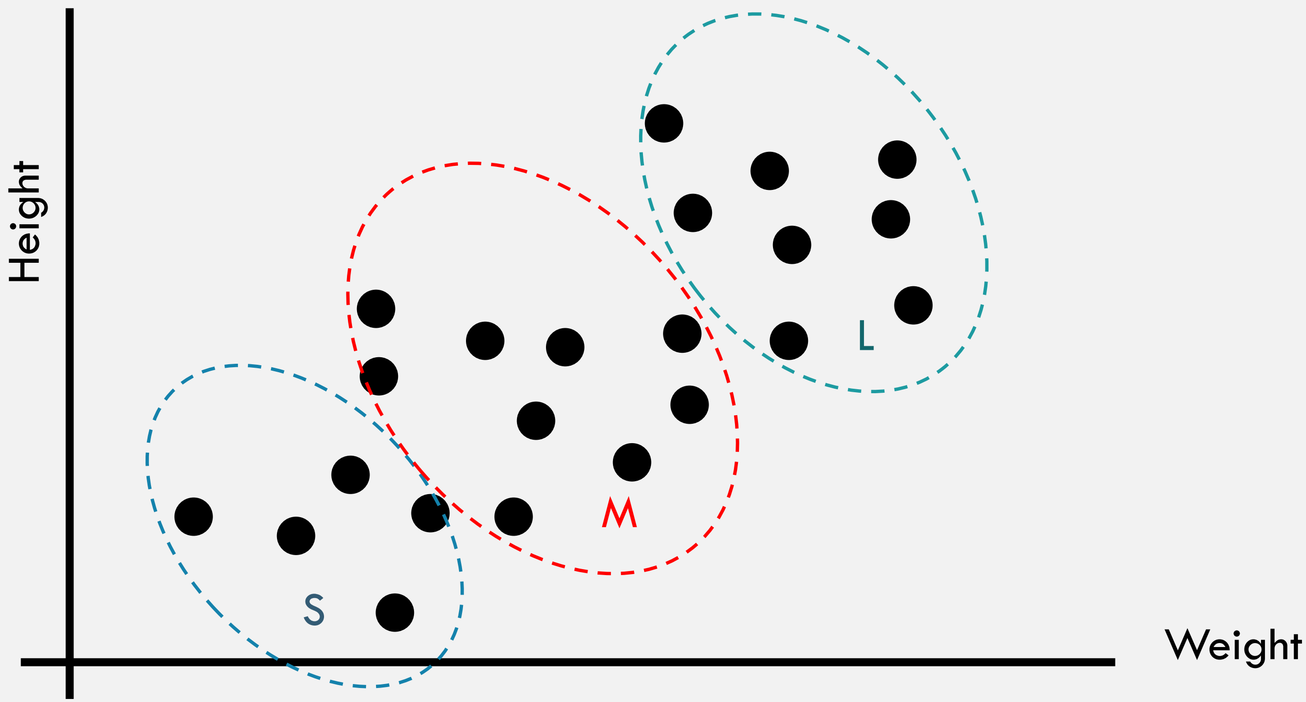


WHAT IS THE BEST K IN THIS EXAMPLE?



TWO MANUAL APPROACHES TO CHOOSE K

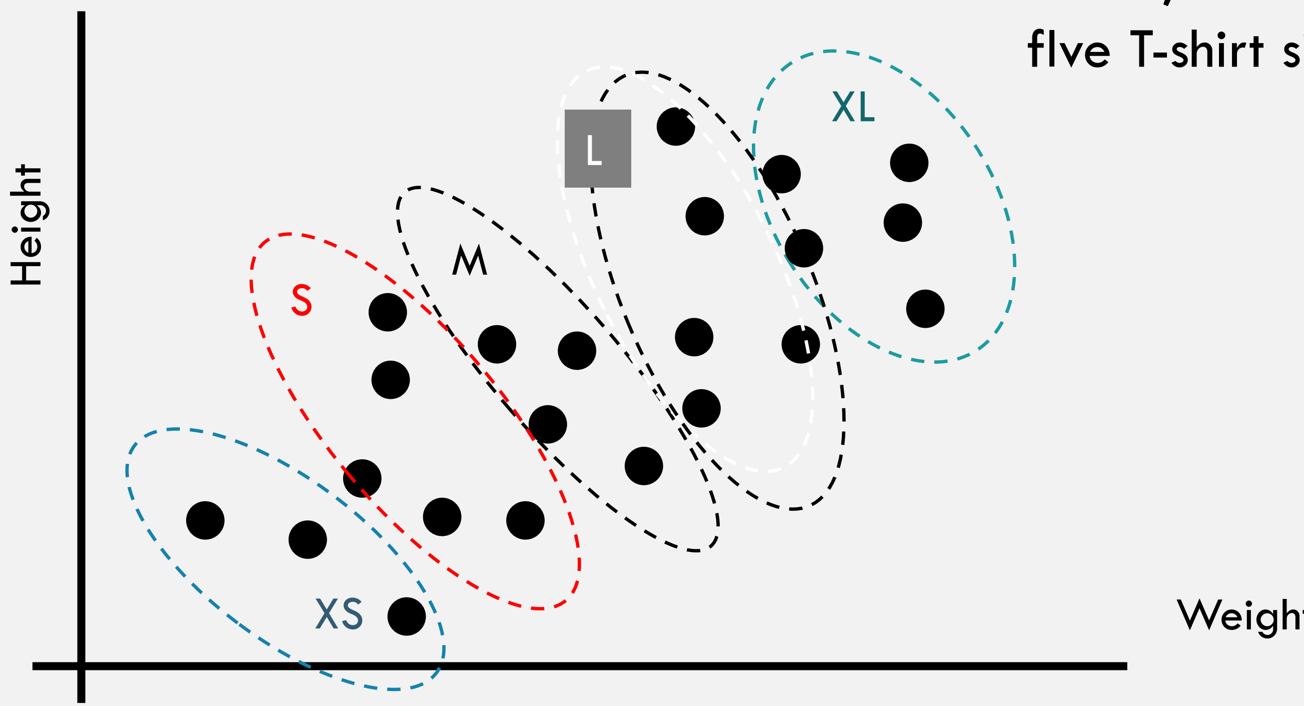
1) Use your intuition and take into consideration the goal of the clustering



TWO MANUAL APPROACHES TO CHOOSE K

1) Use your intuition and take into consideration the goal of the clustering

K = 5,
fIve T-shirt sizes



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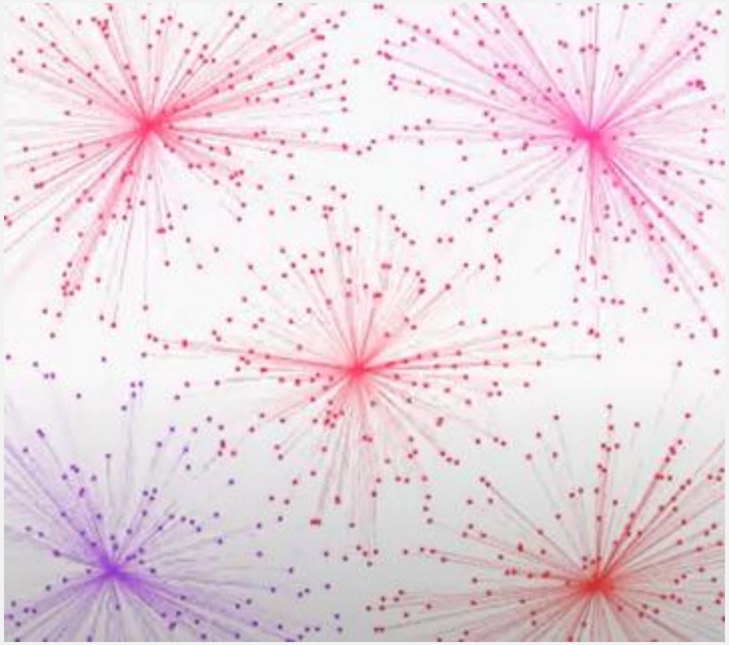
TWO MANUAL APPROACHES TO CHOOSE K

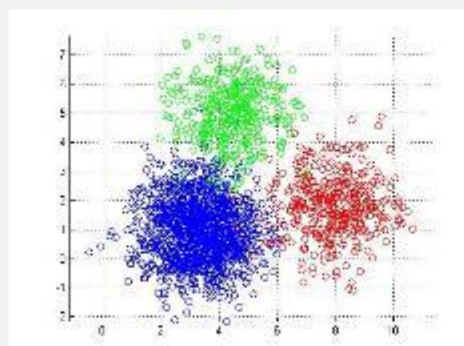
2) Use the elbow method (what is the graph's shape?)

DEMONSTRATION



K–MEANS SUMMARY

- K-means is a very intuitive yet powerful algorithm for unsupervised clustering
- It is efficient in well established clusters but also good for non-separated data
- The algorithm is not deterministic and therefore should be performed several times.
- Choosing the K is important and often must be manual
- Q



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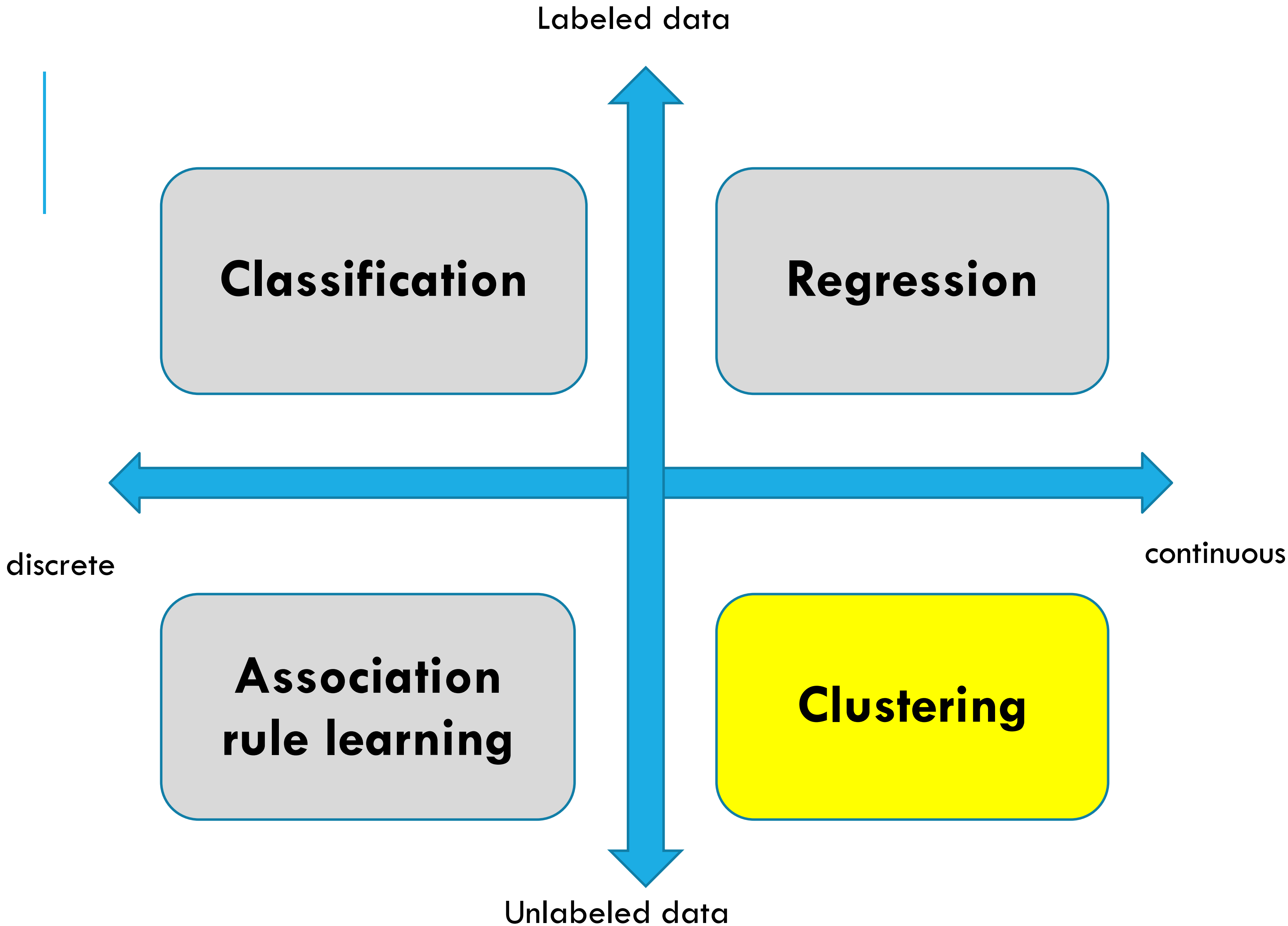
HIERARCHICAL CLUSTERING


Labeled data
Classification
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WHAT IS HIERARCHICAL CLUSTERING?

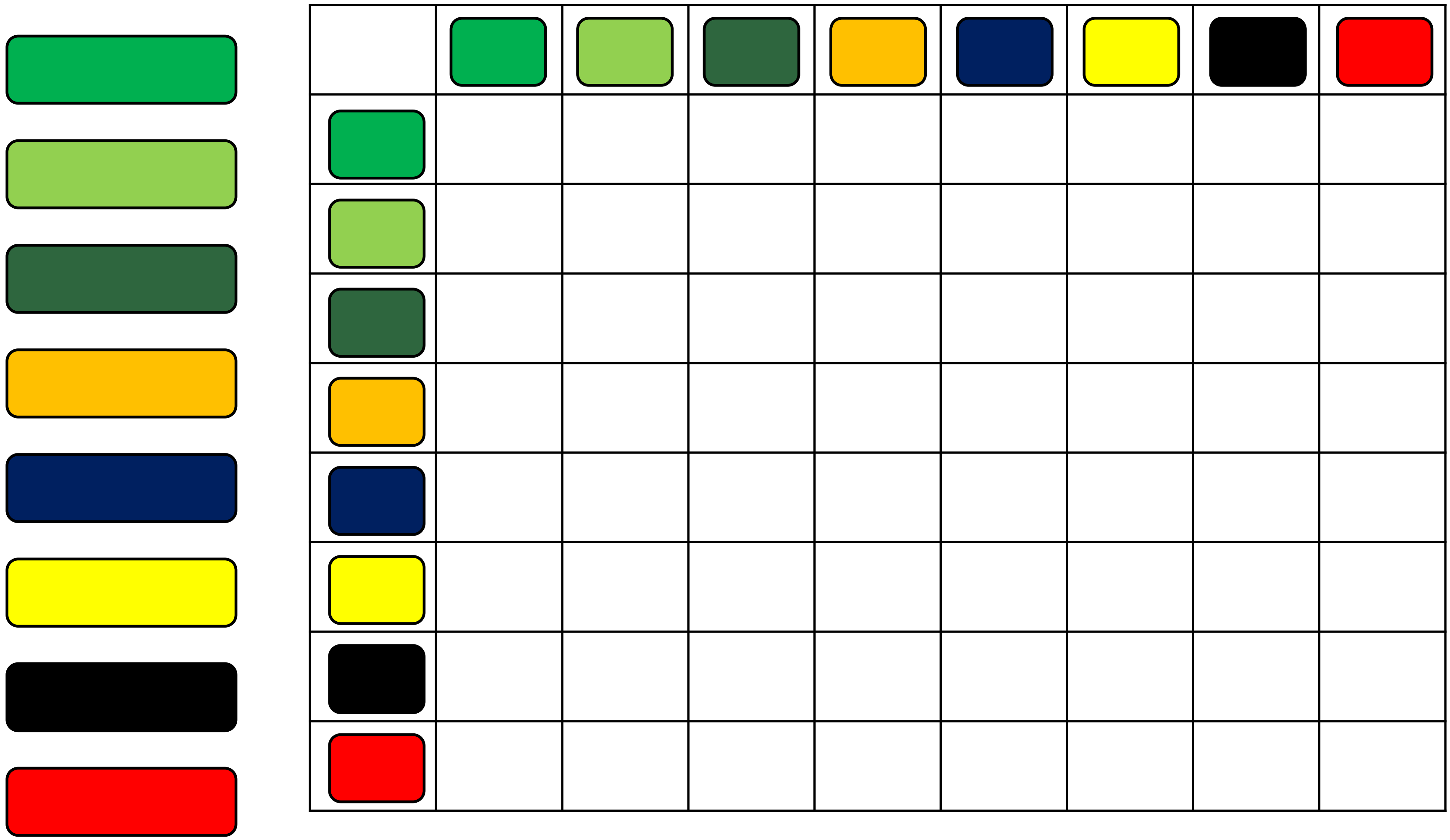
- Order the sample in a hierarchy
- This yields:
    a) samples sorted nicely
    b) a tree-like structure – this is called a dendrogram

- Similar to K-means, the hierarchical clustering:
    a. is also based on a similarity metric.
    b. is an iterative algorithm

HIERARCHICAL CLUSTERING ALGORITHM

- Step 1: find the distances between all samples
- Step 2: iterations:
  - Join two objects to a new node
  - Update the similarity matrix
- The process is repeated m -1 times until only a single element remains

HIERARCHICAL CLUSTERING EXAMPLE – SIMILARITY



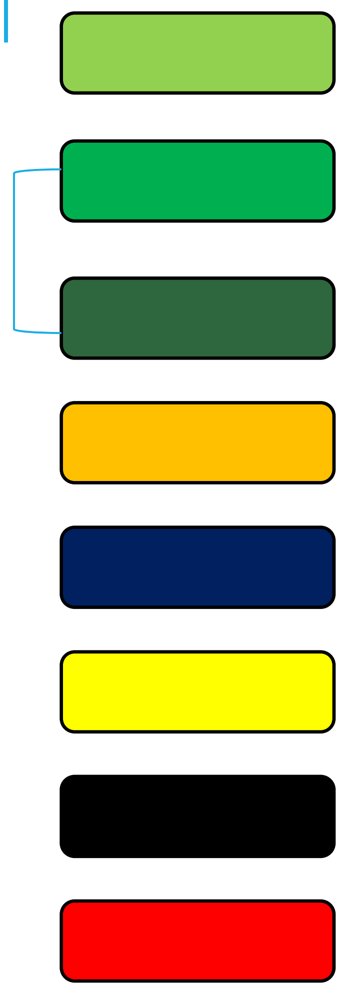
HIERARCHICAL CLUSTERING EXAMPLE

	1							
		1						
			1					
				1				
					1			
						1		
							1	
								1

HIERARCHICAL CLUSTERING EXAMPLE

	1	0.6	0.9	0.3	0.75	0.2	0.7	0.5
		1	0.88	0.2	0.6	0.7	0.2	0.1
			1	0.2	0.77	0.5	0.4	0.2
				1	0.1	0.82	0.12	0.7
					1	0.12	0.65	0.4
						1	0.1	0.4
							1	0.3
								1

HIERARCHICAL CLUSTERING EXAMPLE



	1	0.6	0.9	0.3	0.75	0.2	0.7	0.5
		1	0.88	0.2	0.6	0.7	0.2	0.1
			1	0.2	0.77	0.5	0.4	0.2
				1	0.1	0.82	0.12	0.7
					1	0.12	0.65	0.4
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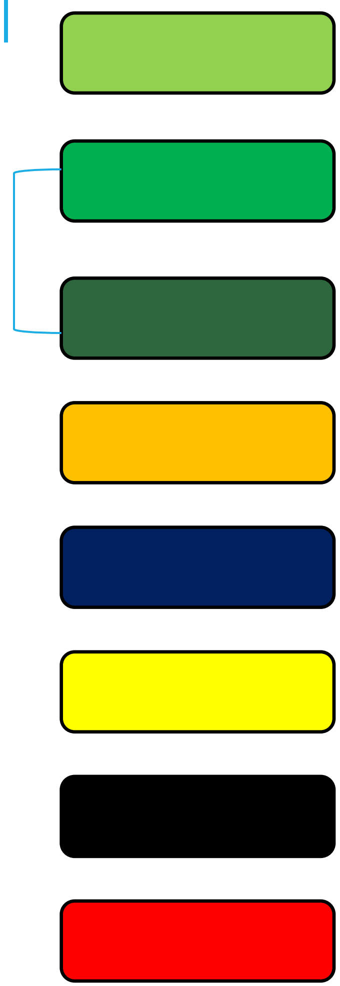
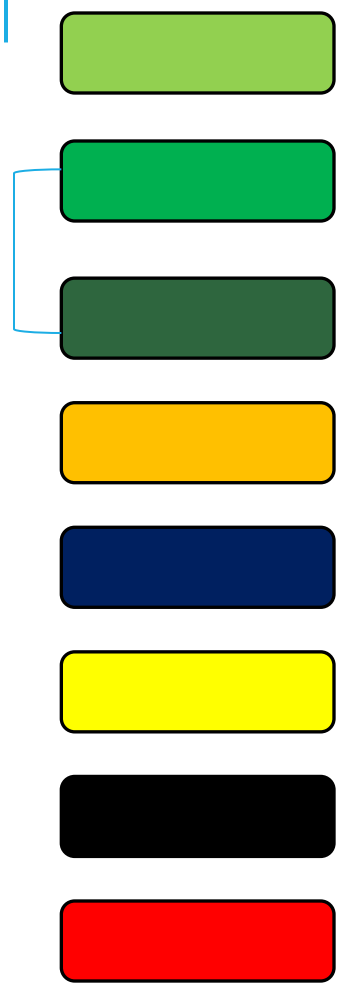


	1						
		1					
			1				
				1			
					1		
						1	
							1

HIERARCHICAL CLUSTERING EXAMPLE



	1	0.74	0.25	0.71	0.35	0.55	0.35
		1	0.2	0.6	0.7	0.2	0.1
			1	0.1	0.82	0.12	0.7
				1	0.12	0.65	0.4
					1	0.1	0.4
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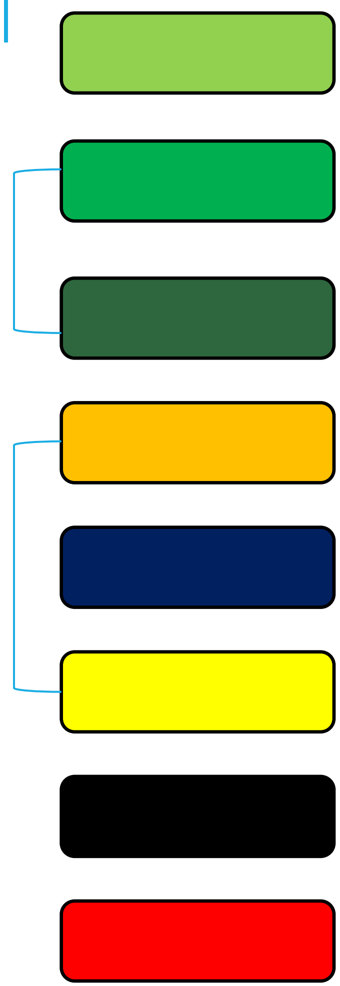
							1

HIERARCHICAL CLUSTERING EXAMPLE



	1	0.74	0.25	0.71	0.35	0.55	0.35
		1	0.2	0.6	0.7	0.2	0.1
			1	0.1	0.82	0.12	0.7
				1	0.12	0.65	0.4
					1	0.1	0.4
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							1

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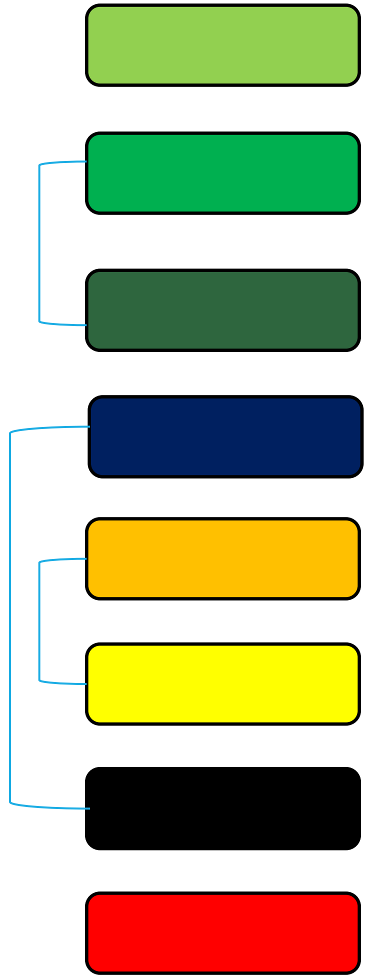

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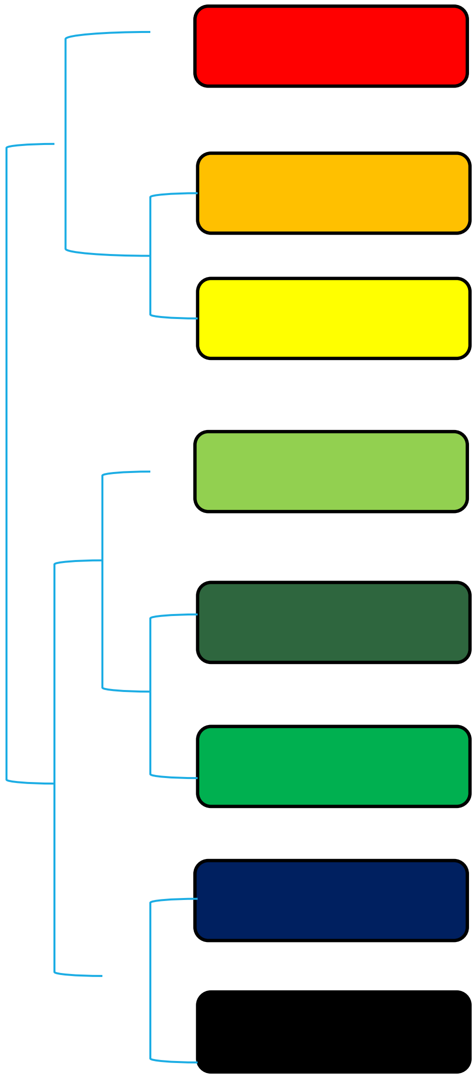
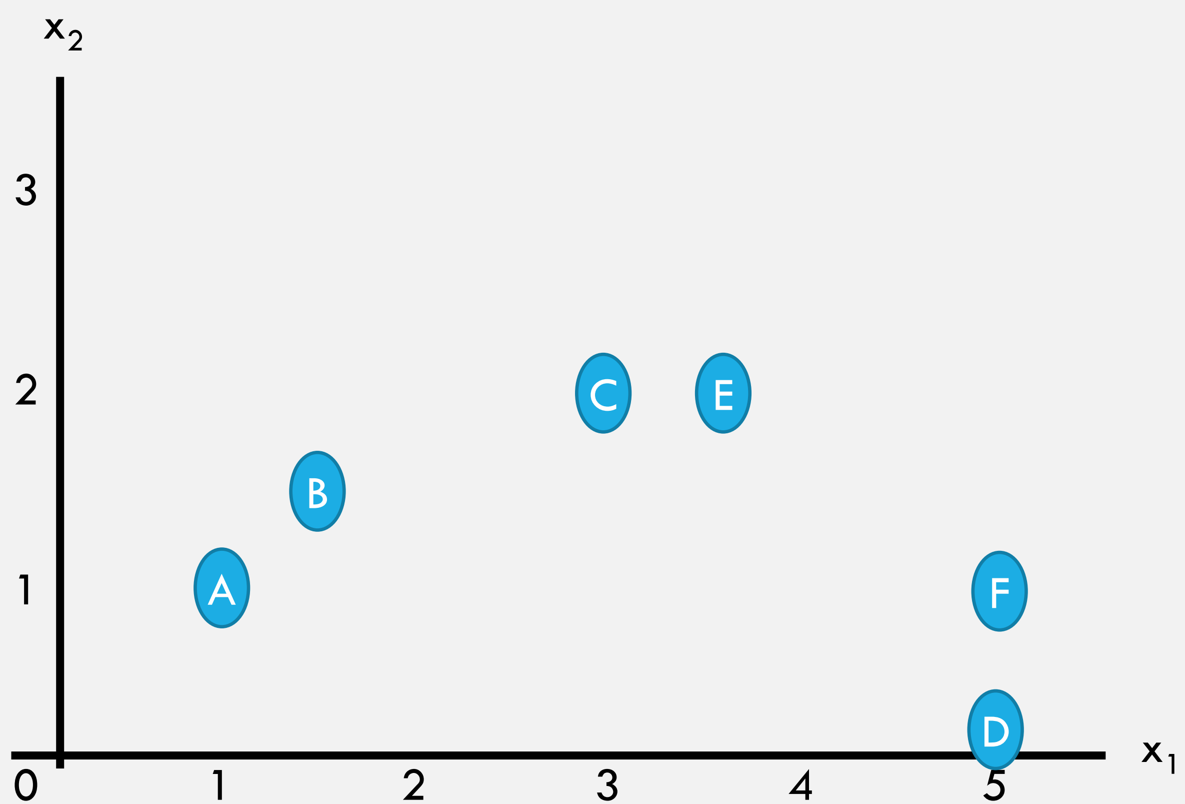


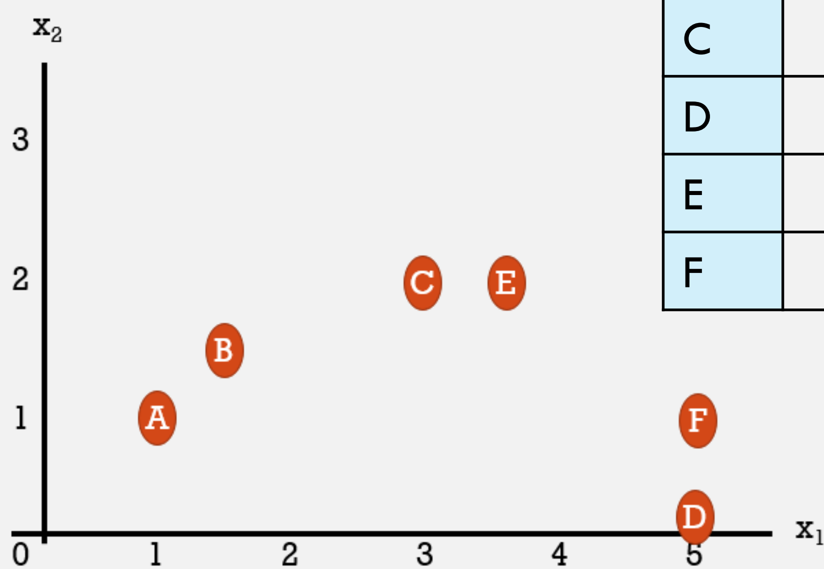
HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE



HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE

Step 1: calculate the distance metrics based on Euclidean distance:

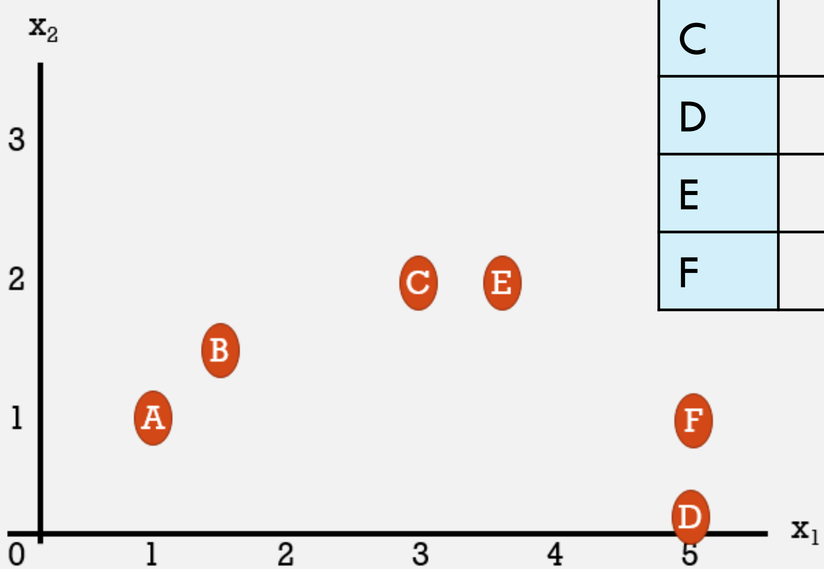
	A	B	C	D	E	F
A						
B						
C						
D						
E						
F						



HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE

Step 1: calculate the distance metrics based on Euclidean distance:

$D_{A,B} = \sqrt{(1.5-1)^2 + (1.5-1)^2}$
$= 0.71$



	A	B	C	D	E	F
A	0					
B		0				
C			0			
D				0		
E					0	
F						0

HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE

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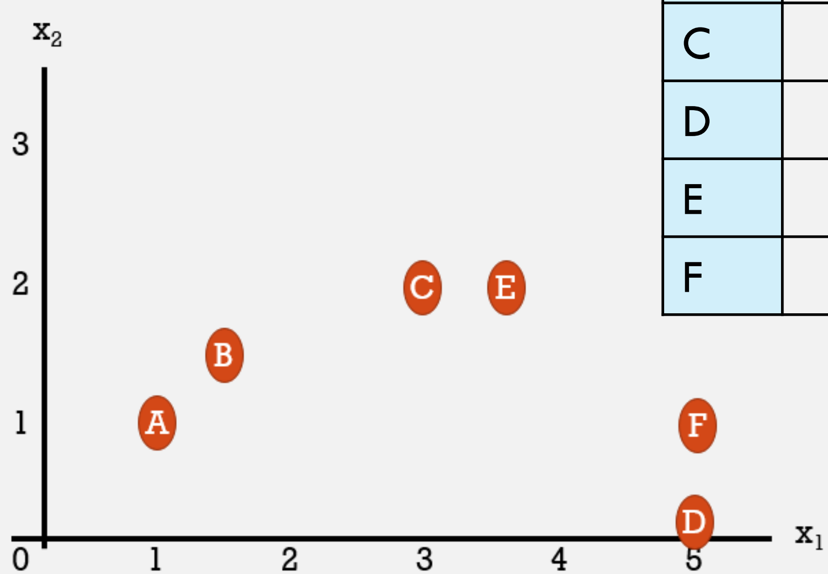
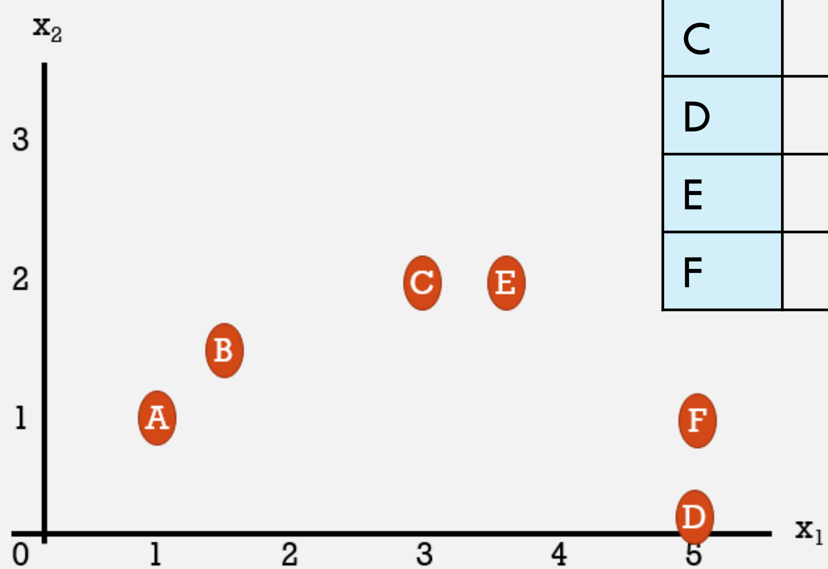


	A	B	C	D	E	F
A	0	0.7				
B		0				
C			0			
D				0		
E					0	
F						0

HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE

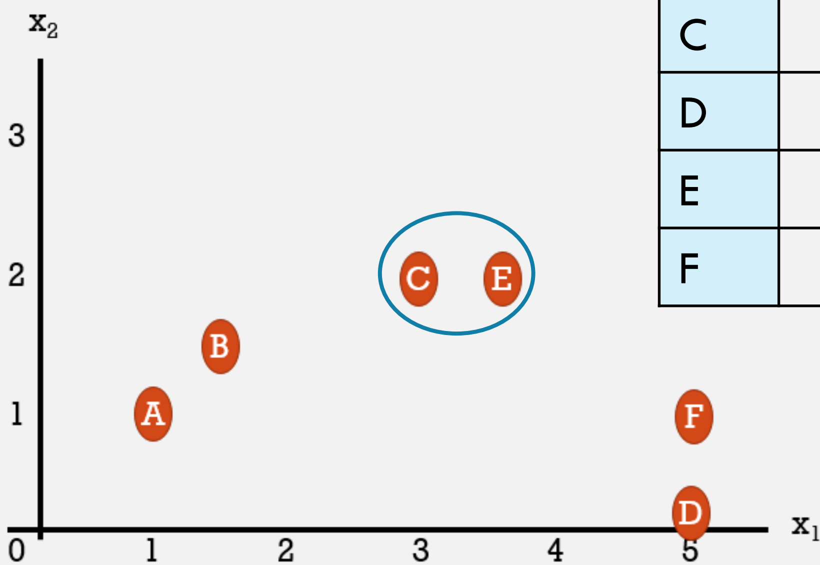
Step 1: calculate the distance metrics based on Euclidean distance:



	A	B	C	D	E	F
A	0	0.7	2.2	4.1	2.7	4
B		0	1.6	3.8	2.1	3.5
C			0	2.8	0.5	2.2
D				0	2.5	1
E					0	1.8
F						0

HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE

Step 2: Find the closest nodes

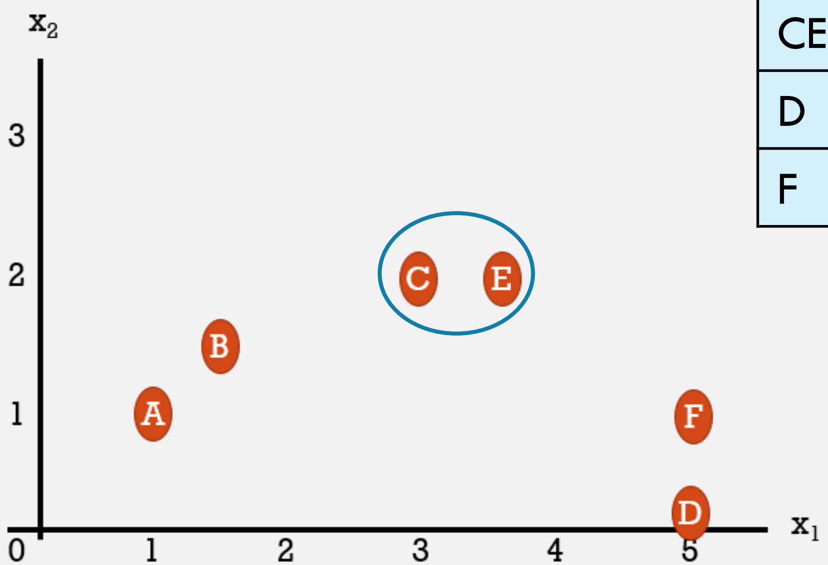


	A	B	C	D	E	F
A	0	0.7	2.2	4.1	2.7	4
B		0	1.6	3.8	2.1	3.5
C			0	2.8	0.5	2.2
D				0	2.5	1
E					0	1.8
F						0

HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE

Merge the two nodes into one

	A	B	CE	D	F
A	0	0.7		4.1	4
B		0		3.8	3.5
CE			0		
D				0	1
F					0



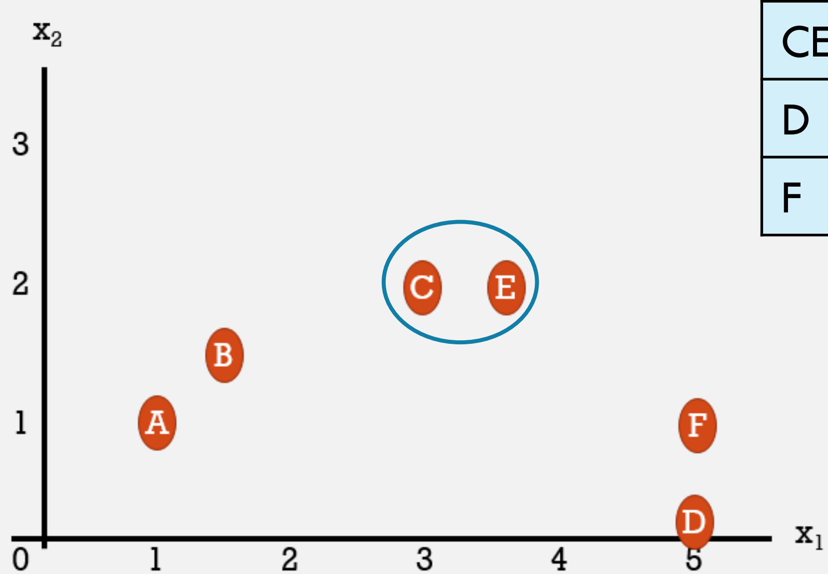
HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE

Merge the two nodes into one

Define their new "position"

(average = (3.25,2))

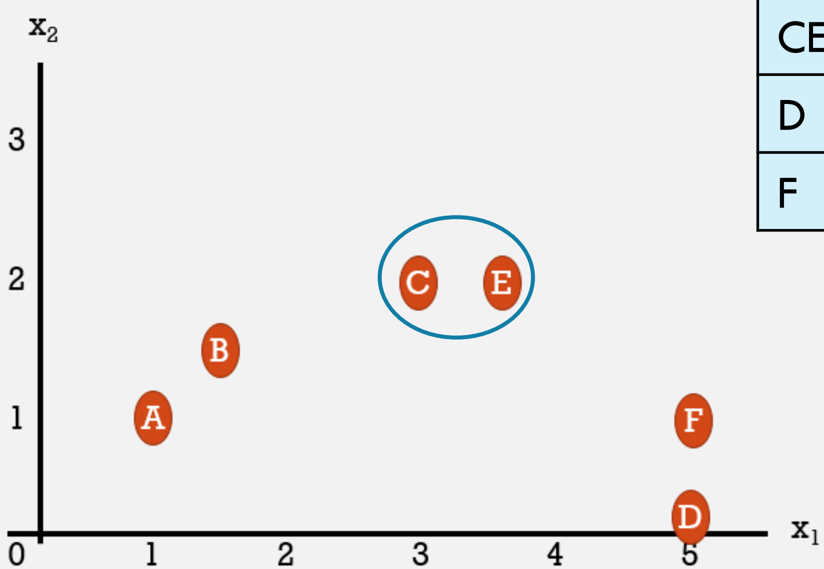
	A	B	CE	D	F
A	0	0.7		4.1	4
B		0		3.8	3.5
CE			0		
D				0	1
F					0



HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE

Re-calculate the matrix

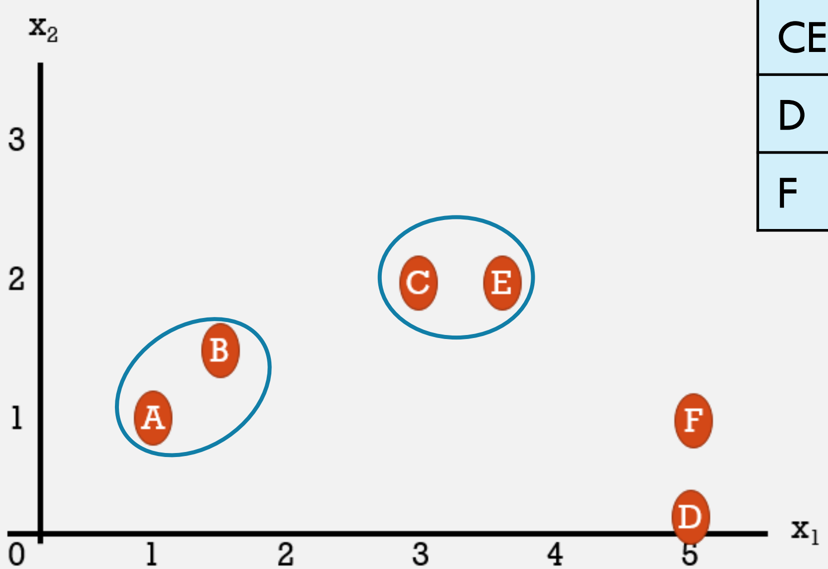
	A	B	CE	D	F
A	0	0.7	2.5	4.1	4
B		0	1.8	3.8	3.5
CE			0	2.7	2.0
D				0	1
F					0



HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE

Find the closest nodes

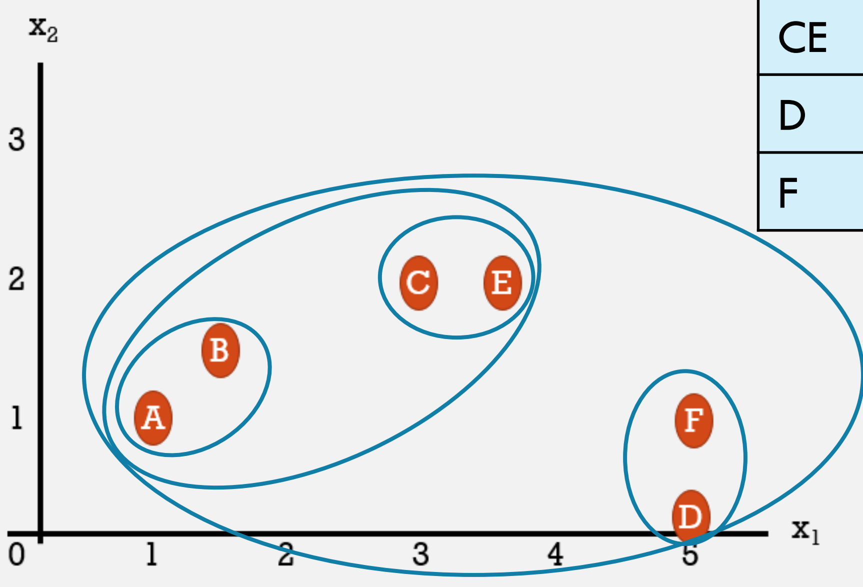
	A	B	CE	D	F
A	0	0.7	2.5	4.1	4
B		0	1.8	3.8	3.5
CE			0	2.7	2.0
D				0	1
F					0



HIERARCHICAL CLUSTERING BY DISTANCE EXAMPLE

And so on…

	A	B	CE	D	F
A	0	0.7	2.5	4.1	4
B		0	1.8	3.8	3.5
CE			0	2.7	2.0
D				0	1
F					0



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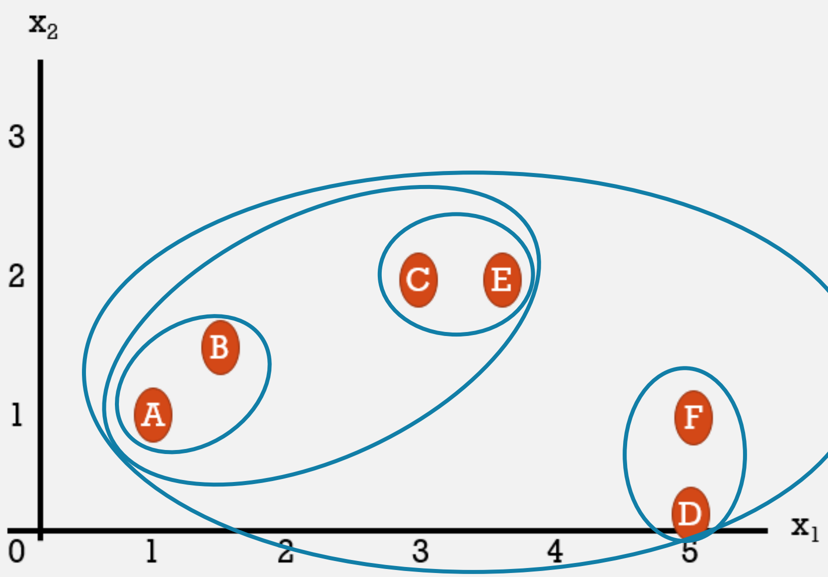
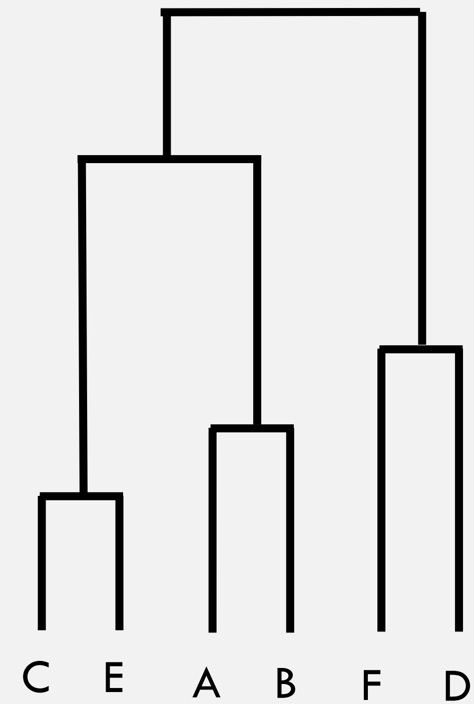
NOW LET'S BUILD THE DENDROGRAM

- We started with CE
- Then AB
- FD
- AB-CE
- Overall cluster





Distances are represented graphically

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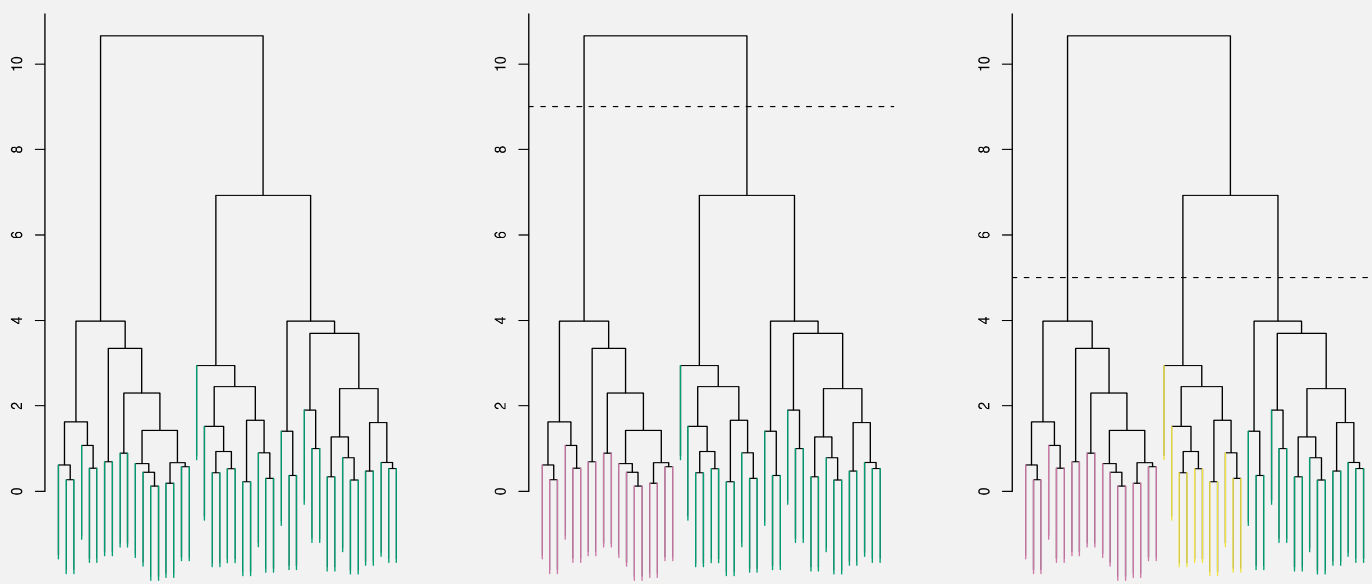
CHOOSING CLUSTERS

To choose clusters we cut the dendrogram at a specific height

We can form any number of clusters depending on the cut height



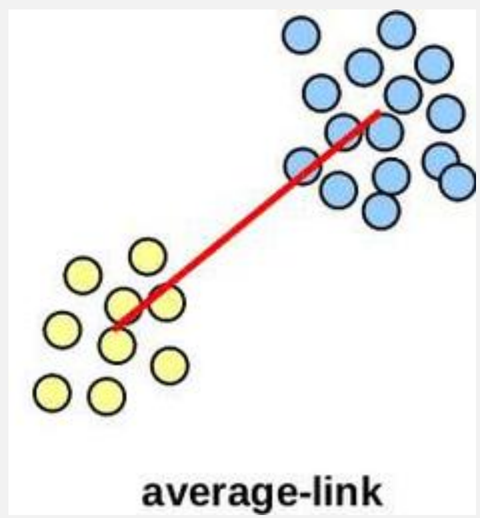
LINKAGE IN HIERARCHICAL CLUSTERING

- We already know about distance measures between data items
- But how can we calculate distance measures between
  - a data item and a cluster
  - between two clusters?
- We just treat a cluster as a single data point, so our only problem is to define a ***linkage*** method between clusters
- As usual, there are lots of choices…

AVERAGE LINKAGE

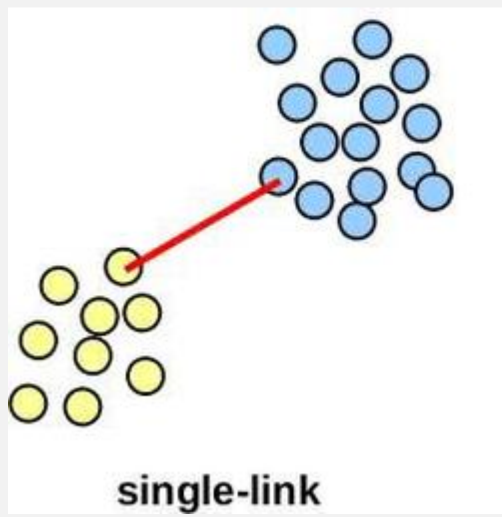
The **average** of all pairwise distances between points in the two clusters

- Note! It's not the distance between the clusters' means - Centroid Linkage
- Tends to produce evenly sized clusters



SINGLE LINKAGE

- The **minimum** of all pairwise distances between points in the two clusters
- Tends to produce long, “loose” clusters



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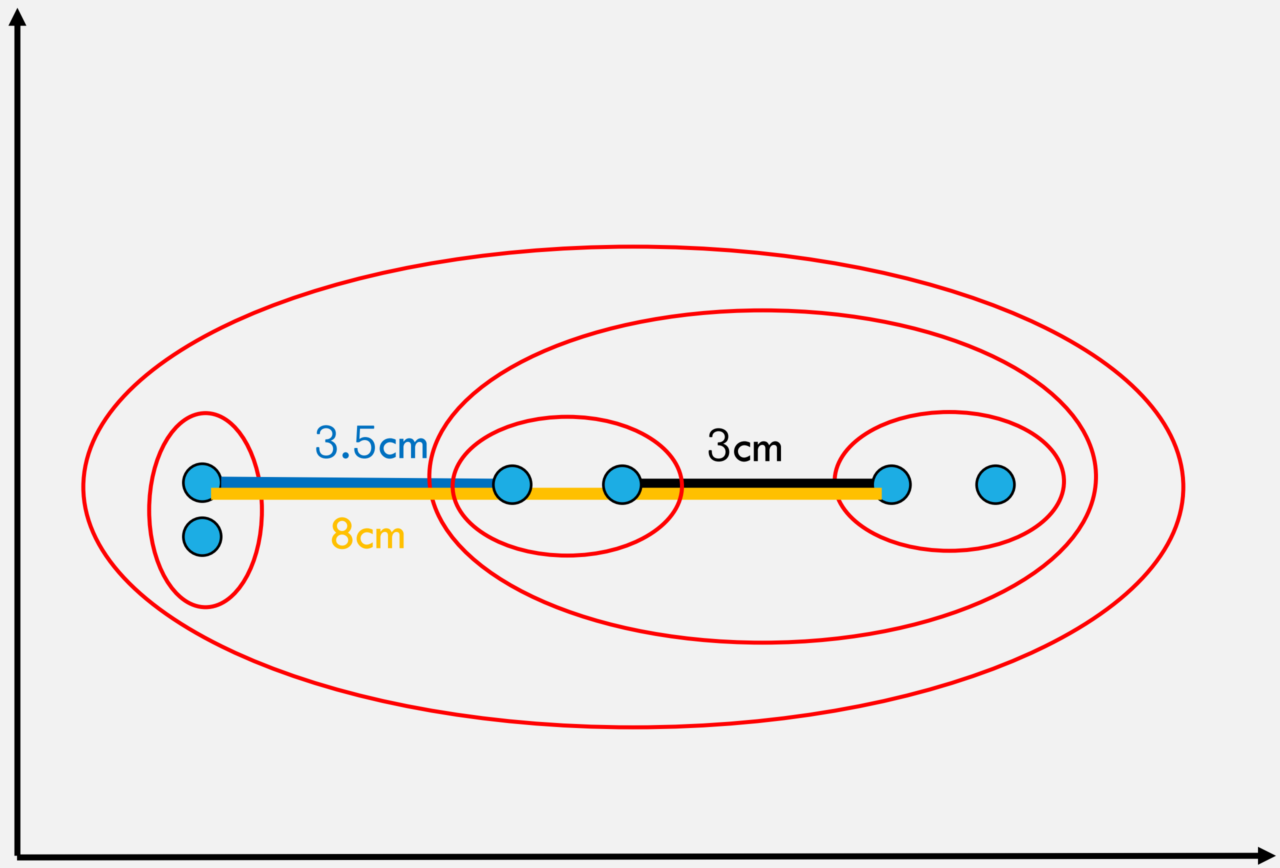
SINGLE LINKAGE EXAMPLE



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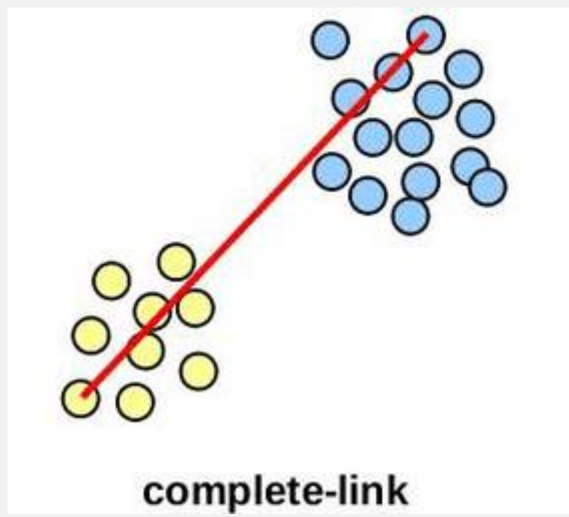
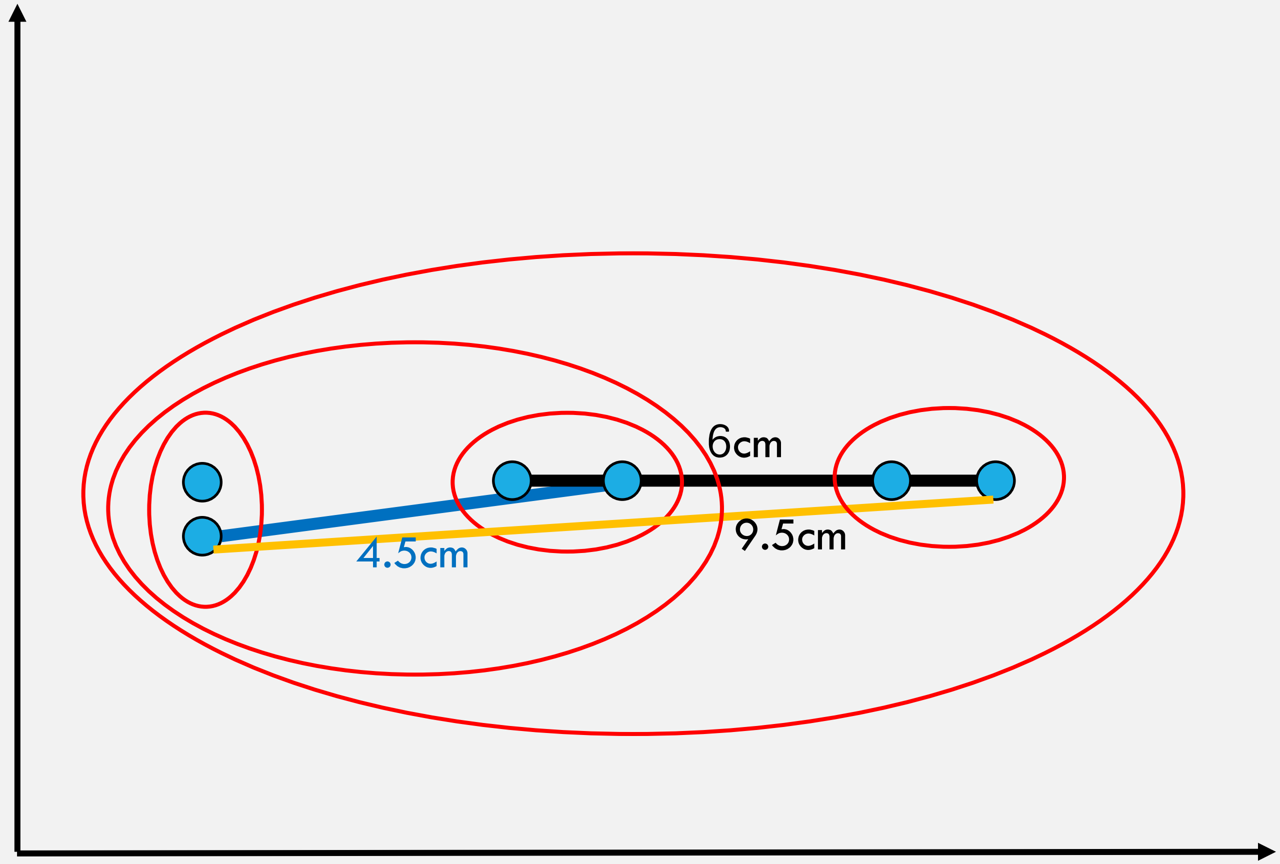
COMPLETE LINKAGE

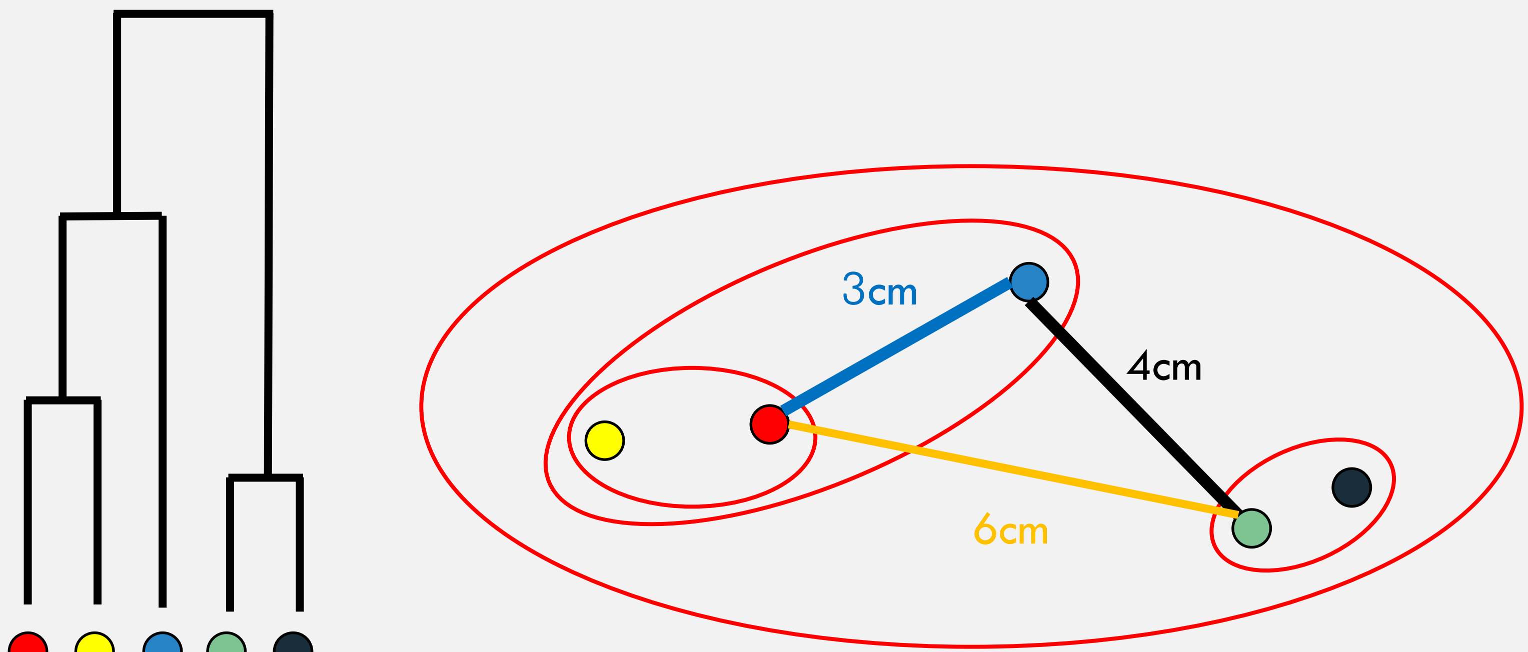
- The **maximum** of all pairwise distances between points in the two clusters
- Tends to break large clusters



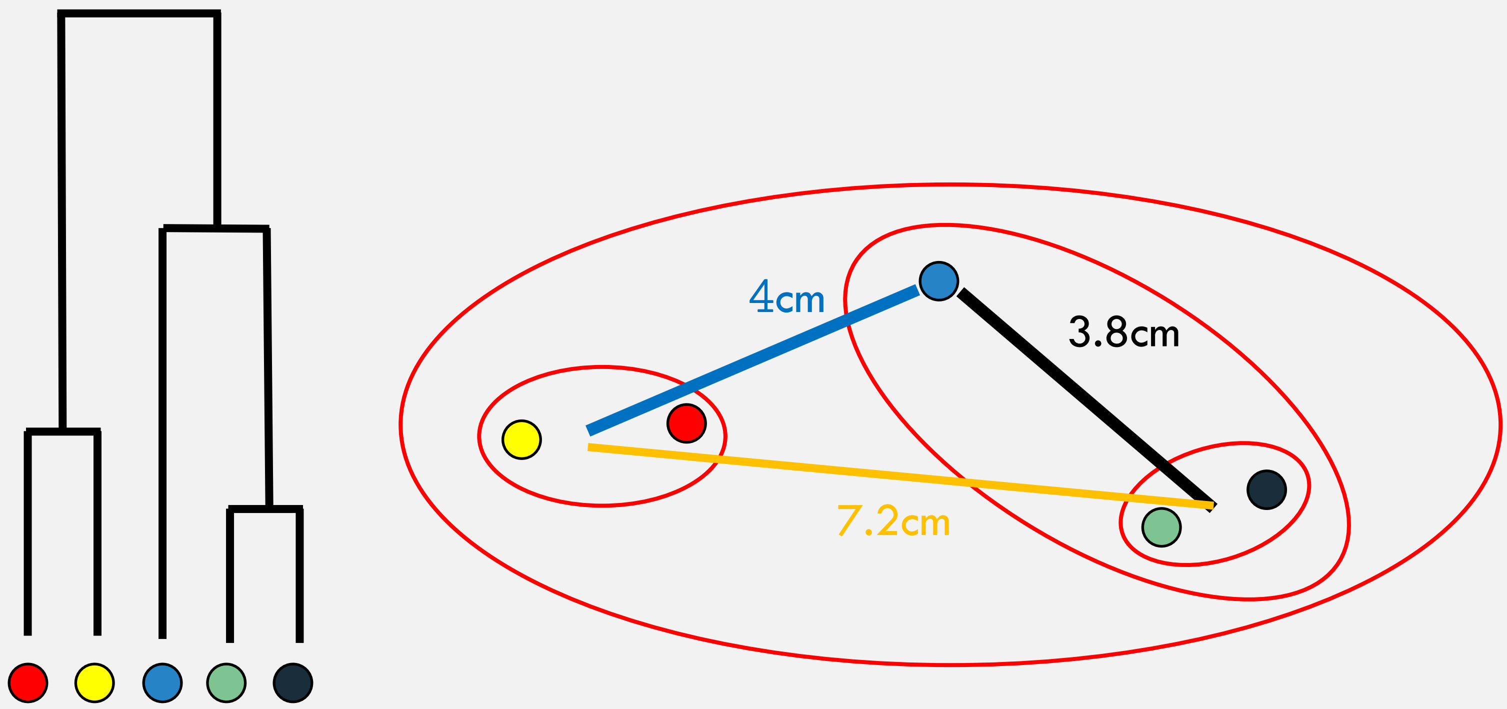
COMPLETE LINKAGE EXAMPLE



MORE EXAMPLES – SINGLE LINKAGE



MORE EXAMPLES – AVERAGE LINKAGE



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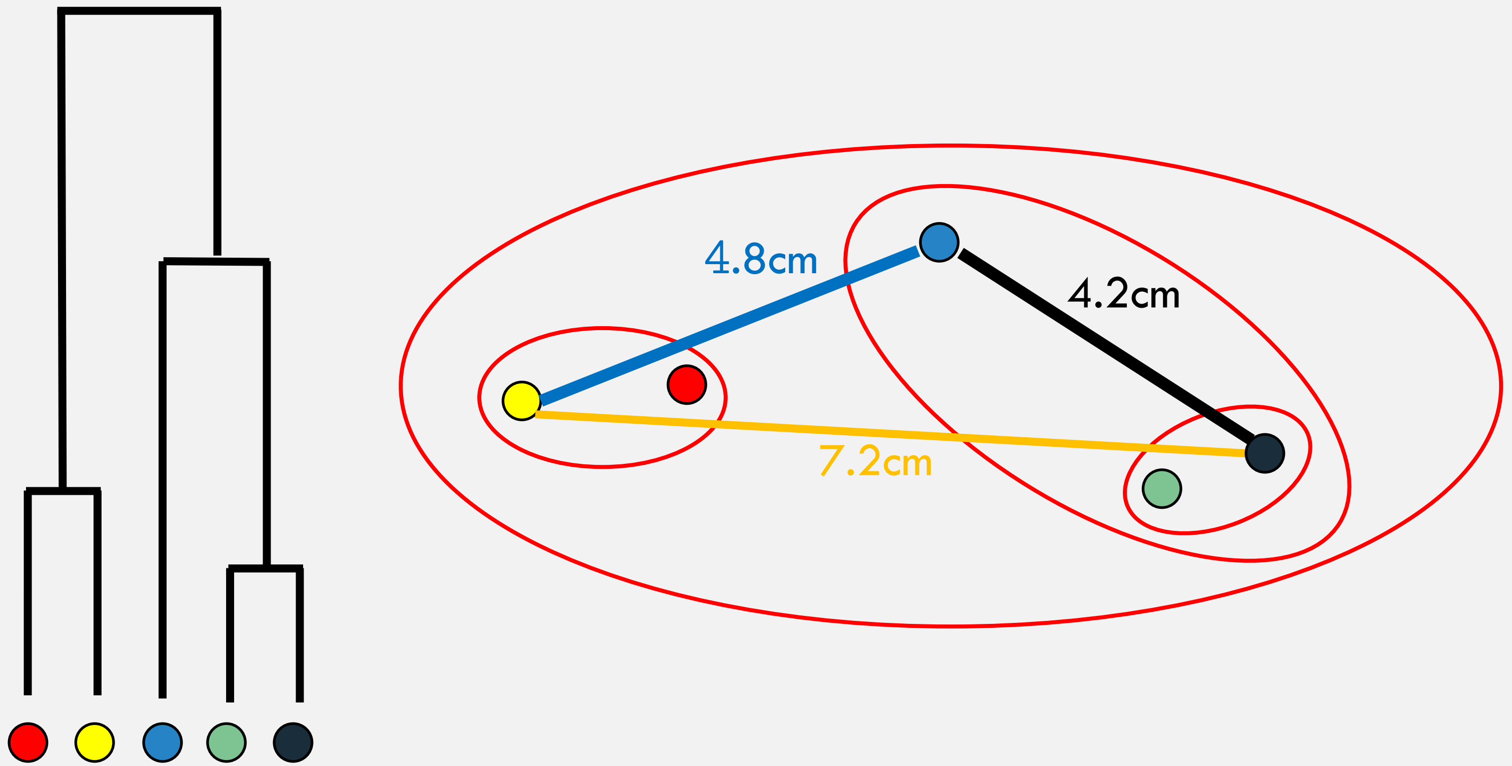
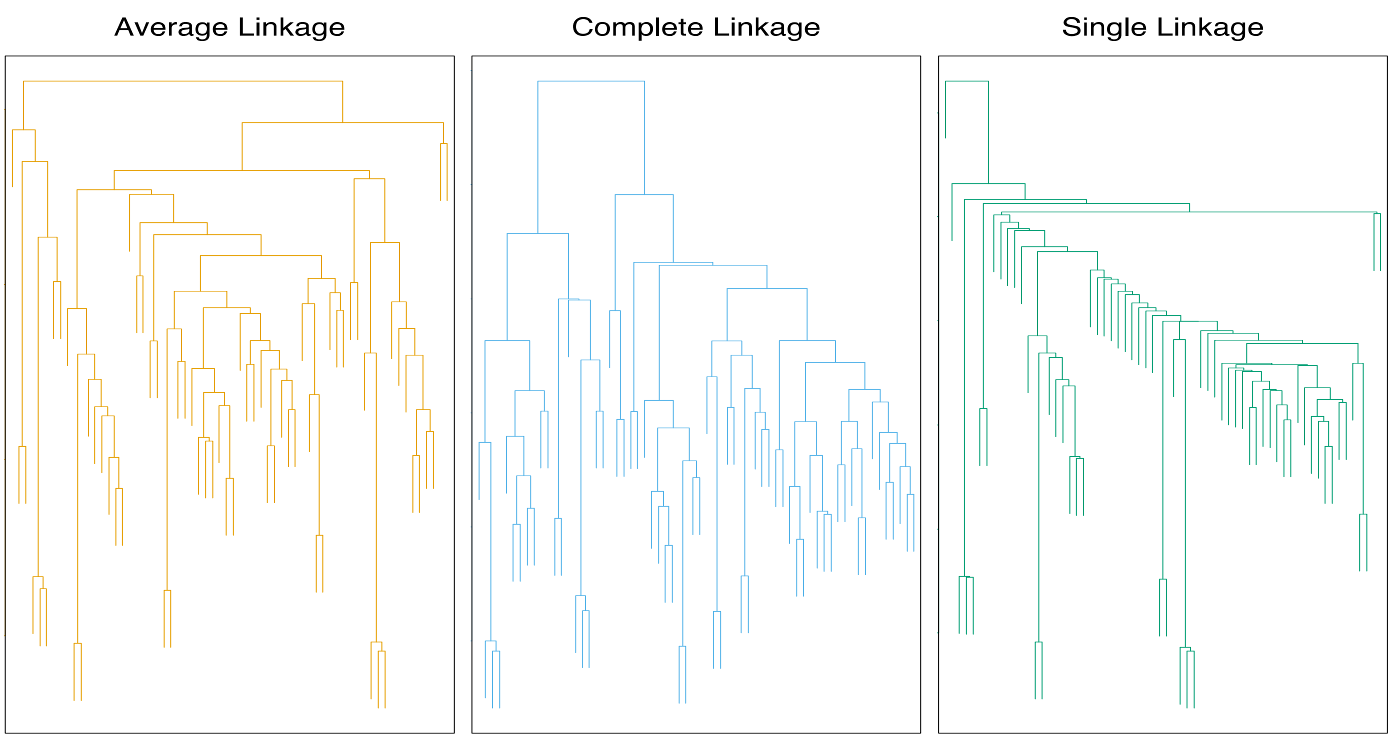
MORE EXAMPLES – COMPLETE LINKAGE



LINKAGE

- Complete and average linkage → evenly sized clusters
- Single linkage → extended clusters to which single leaves are fused one by one.



HIERARCHICAL CLUSTERING – PROS AND CONS

Pros

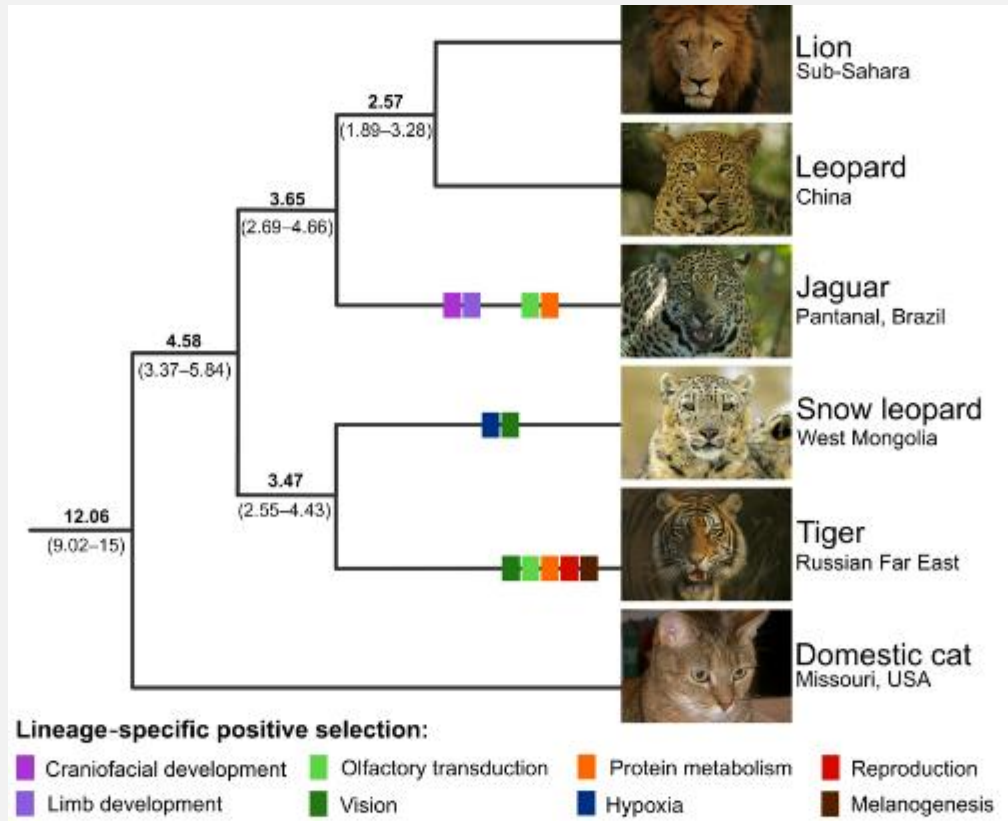
- No need to assume the number of clusters, just cut the tree
- Hierarchy is important and makes sense in many fields
- Informative output that shows the distances between clusters

Cons

- Time consuming, more calculations
- Less trivial: interpreting it and finding where to cut the tree
- Many decisions: distance metric, linkage, where to cut the tree

HIERARCHICAL CLUSTERING – APPLICATIONS

- Build a tree-based hierarchical taxonomy (*dendrogram*) from a set of documents
- Phylogenetic trees



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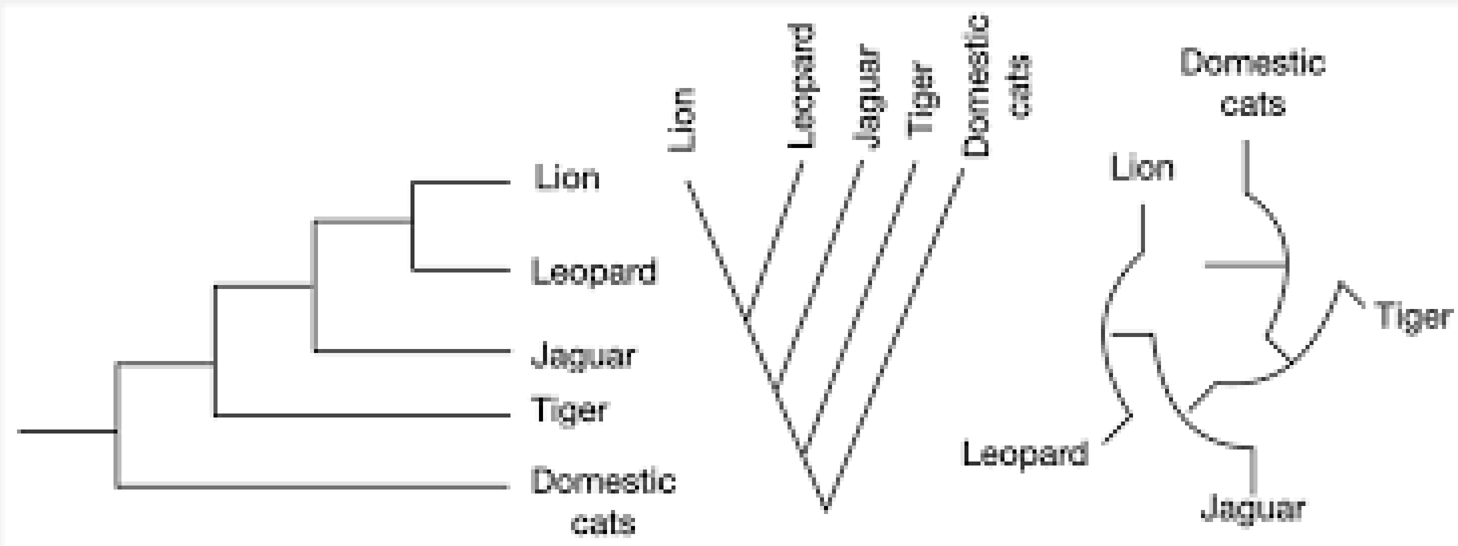
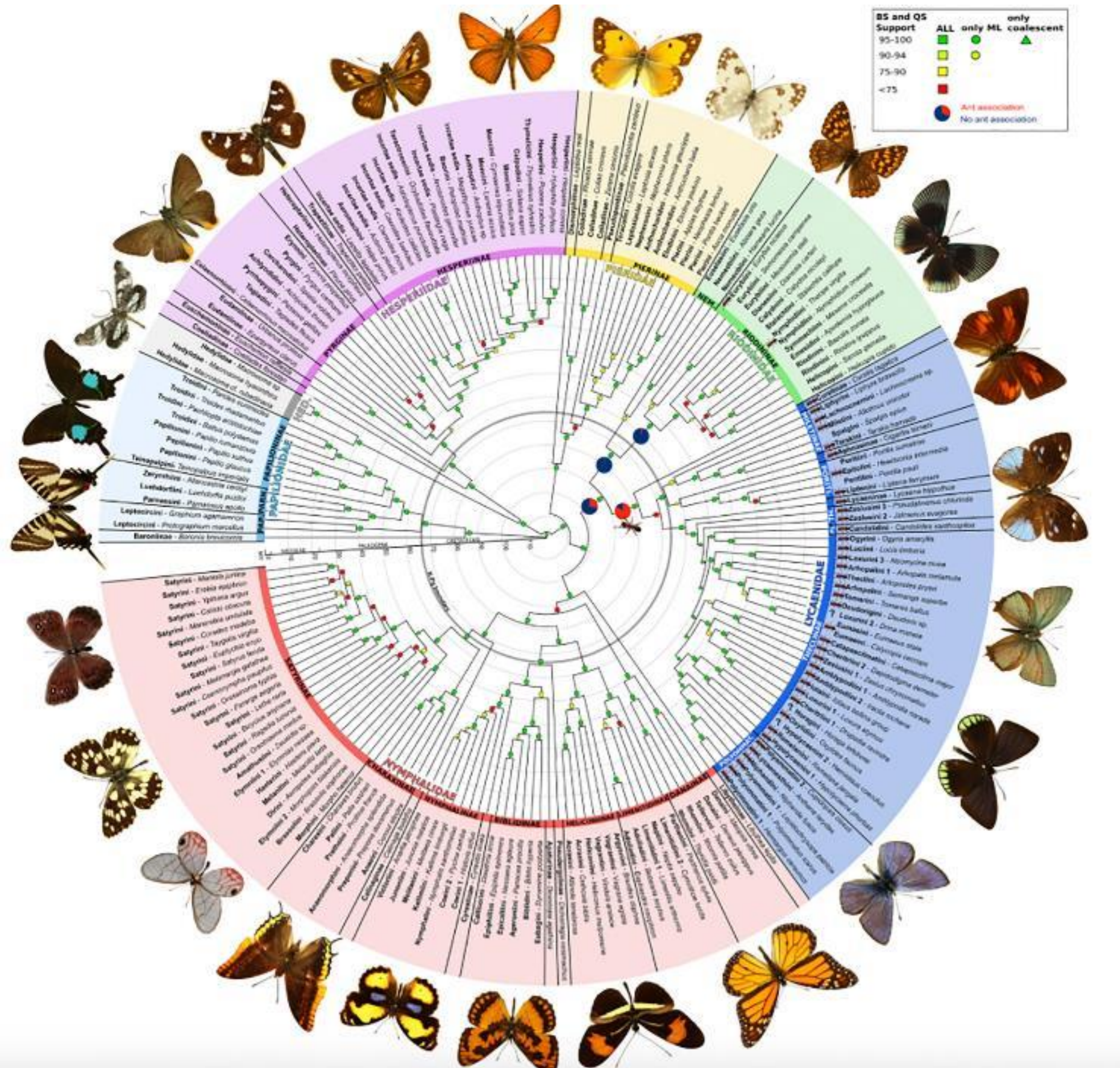
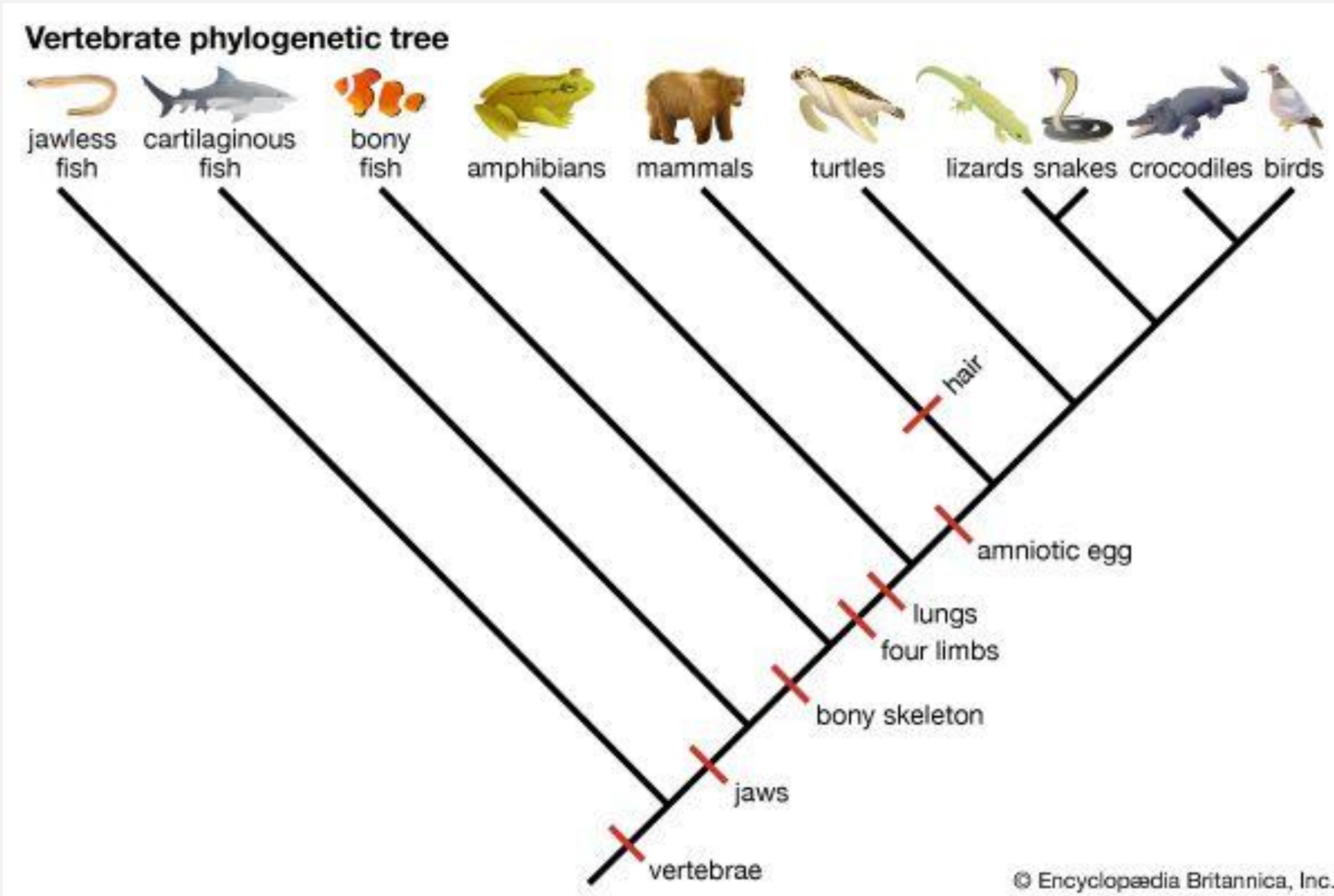
DIFFERENT TYPES OF DENDROGRAMS



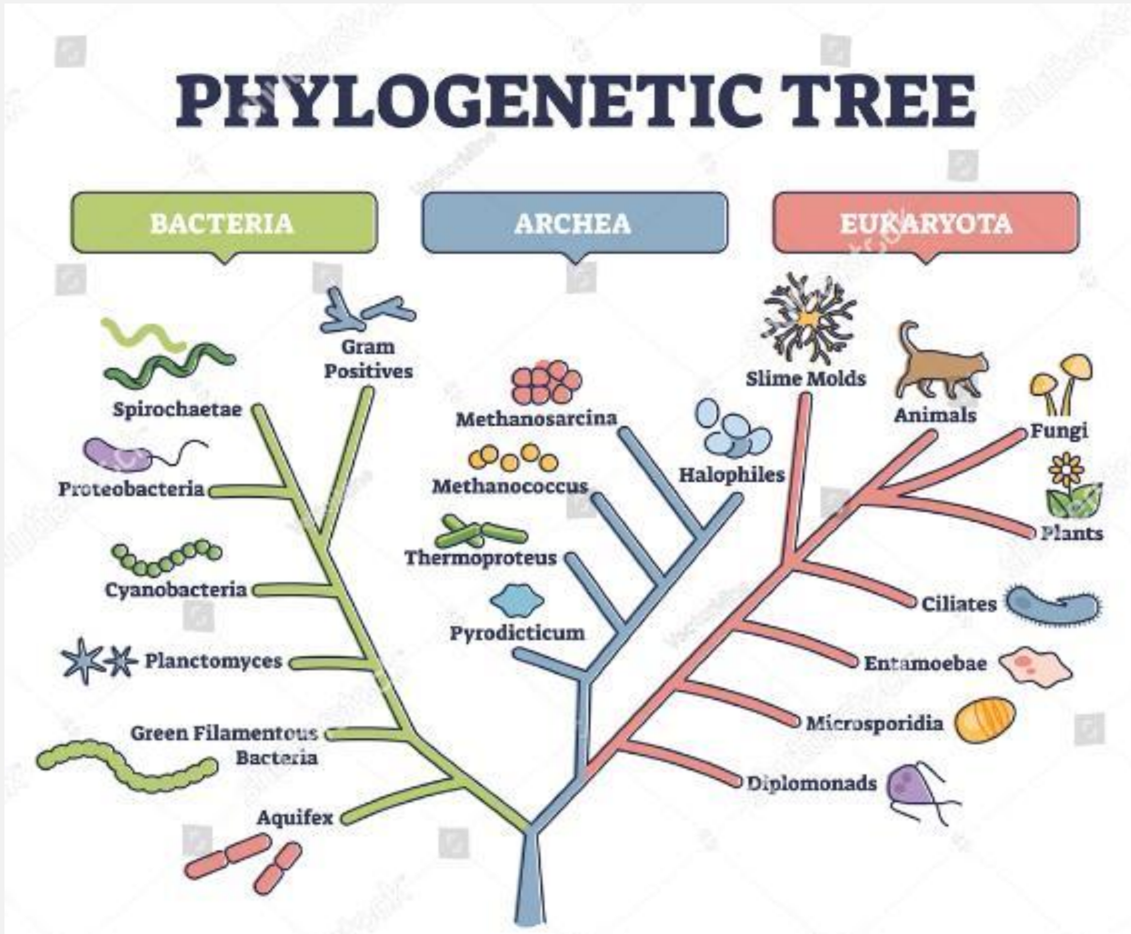
DIFFERENT SHAPES OF DENDROGRAMS



DIFFERENT SHAPES OF DENDROGRAMS



DIFFERENT SHAPES OF DENDROGRAMS



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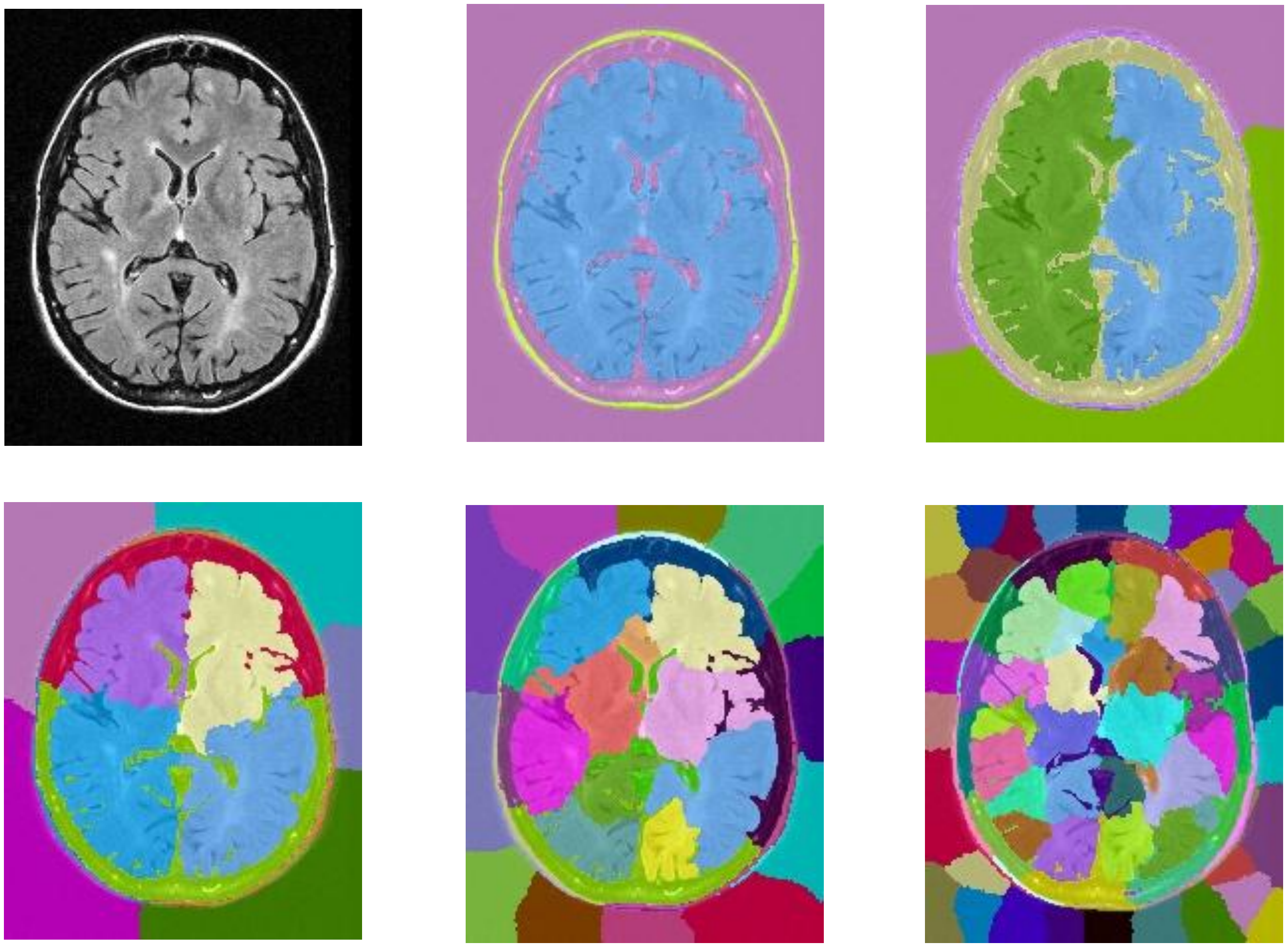
HIERARCHICAL CLUSTERING – APPLICATIONS



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HIERARCHICAL CLUSTERING TAKE HOME MESSAGE

Hierarchical clustering is one of the most efficient and useful clustering algorithm

It relies on iterations in which we choose the closer nodes while building a dendrogram

Cutting the dendrogram enable segmentation of the data into the clusters

Choosing the right linkage approach is important for an appropriate clustering